



Striving for balance: A look at gameplay requirements of massively multiplayer online role-playing games

Maya Daneva

Computer Science Department, University of Twente, Drienerloaan 5, Enschede The Netherlands



ARTICLE INFO

Article history:

Received 17 June 2016

Revised 29 May 2017

Accepted 2 August 2017

Available online 3 August 2017

Keywords:

Requirements engineering

Gameplay requirements

Requirements elicitation

Qualitative interview-based study

Empirical research method

ABSTRACT

Engineering gameplay requirements is the most important task for game development organizations. Game industry discourse is concerned with continuous redesign of gameplay to enhance players' experience and boost game's appeal. However, accounts of gameplay requirements practices are rare. In responding to calls for more research into gameplay requirements engineering, we performed an exploratory study in the context of massively multiplayer online role-playing games (MMORPGs), from the perspective of practitioners involved in the field. Sixteen practitioners from three leading MMORPG-producing companies were interviewed and their gameplay requirements documents were reviewed. Interviewing and qualitative data analysis occurred in a cyclical process with results at each stage of the study informing decisions about data collection and analysis in the next.

The analysis revealed a process of striving to reach a balance among three perspectives of gameplay requirements: a process perspective, an artifact perspective and a player-designer relationship perspective. This balance-driven process is co-created by game developers and players, is endless within the MMORPG, and is happening both in-game and off-game. It heavily relies on 'paper-prototyping' and play-testing for the purpose of gameplay requirements validation. The study concludes with discussion on validity threats and on implications for requirements engineering research, practice and education.

© 2017 Elsevier Inc. All rights reserved.

1. Introduction

In the past ten years, companies producing massively multiplayer online role-playing games (MMORPGs), such as World of Warcraft, Lineage or EverQuest, have emerged as the fastest growing industry, generating worldwide revenue expected to reach 42 billion US\$ by 2020 (DFC Intelligence 2016; De Heij, 2013). Essentially, such games are entertainment software system (of systems) providing 3D richly detailed online environments to millions of geographically distributed players to simultaneously participate in game communities and engage in social interaction and game activities, while exploring an unfriendly fantasy world. Vast and complex, MMORPGs result from long and expensive game design processes (DFC Intelligence 2016; De Heij, 2013). The focal point in such processes is the so-called 'gameplay' which is most generally regarded in MMORPG literature (e.g. Nardi, 2010) as to what players and non-player characters (NPCs) in the game do that is entertaining (also known as 'fun'). For example, the gameplay of Sony's MMORPG EverQuest is collaboratively planning and executing missions to defeat monsters. Gameplay as part of MMORPG design is a subject of both active debate in business and schol-

ars' circles. Professional institutions such as the International Game Developers Organization (IGDA, www.igda.org), the Digital Game research Association (DiGra) and the Entertainment Software Association seek to establish 'good practice' standards to help define and analyze MMORPG gameplay. Similarly, empirical research scholars in the fields of game design (e.g. Fullerton, 2008; LeBlanc, 2006; Zagal and Bruckman, 2008; Bjoerk and Holopainen, 2005; Falstein, 2004), human-computer interaction (e.g. Deaker, 2012; Gross, 2012; Bergstroem, 2010), social behavior (e.g. Nardi, 2010), media communication (e.g. Ducheneaut, 2006), psychology (e.g. Brockmyer, 2009) and education (e.g. Dickey, 2011), have been actively investigating gameplay from the specific theoretical perspectives pertaining to their respective disciplines. Such studies investigated, among other topics, the aspects of gameplay critical to achieving MMORPG designs with mass appeal, the motivational factors concerning players' participation in MMORPG worlds, the MMORPG community-building patterns, and aspects of gameplay critical to human learning. Despite the wealth of gameplay-related research, gameplay only relatively recently became a topic of active research in the discipline of Requirements Engineering (RE) (e.g. Callele, 2005; Alves, 2007; Pasquale, 2013; Cooper, 2014; Kasurinen, 2014). Research on gameplay requirements is relatively rare and whatever empirical evidence it produced, it covered mostly design of standalone video games (e.g.

E-mail address: m.daneva@utwente.nl

Table 1
Related work on RE for games.

Publication	Type of game application	Is gameplay requirements explicitly discussed?
Callele (2005)	Video games	Mentioned only as part of RE for video games
Alves (2007)	Mobile phone games	Mentioned only as part of RE for mobile games
Djaouti et al. (2008)	Video games	Explicit treatment of gameplay requirements
Holopainen (2011)	Video games	Explicit treatment of gameplay requirements
Pasquale (2013)	Console-based games	Mentioned only as part of RE for console games
Cooper (2014)	Serious games	Mentioned only as part of RE for serious games
Kasurinen (2014)	Video games	Explicit treatment of gameplay requirements
Guiard et al. (2014)	Interactive computer games for children with autistic spectrum disorder	Mentioned as part of design processes for games; no discussion on gameplay requirements
Mueller et al. (2014)	Video games	Mentioned as part of design processes for video games; no discussion on gameplay requirements.
Östblad et al. (2014)	Audio-based adventure games for blind or sighted players	Mentioned as part of design processes for games; no discussion on gameplay requirements.
Quax et al. (2014)	MMORPGs	Mentioned as part of MMORPG classification; no discussion on gameplay requirements.
Paschali et al. (2014)	Various genres of computer games, incl. MMORPG	Mentioned only as part of engineering non-functional requirements related to enjoyment factor.
Callele et al. (2015)	Video games	Mentioned only as part of engineering experience requirements and testing those, for video games
Teruel et al. (2016)	Collaborative games, MMORPG included	Mentioned only as part of engineering game space awareness requirements, for collaborative players' games, including MORPG
Scacchi (2017)	Computer games in general	Mentioned only as part of RE for computer games.
Valente et al. (2017)	Mobile games	Mentioned only as part of engineering pervasiveness requirements, for mobile games

Callele, 2005; Pasquale, 2013), console-based (e.g. Cooper, 2014; Kasurinen, 2014) and mobile phone games (e.g. Alves, 2007; Kasurinen, 2014). While the authors of published research on RE for game systems (as the references in Table 1 in Section 2) had acknowledged the paramount importance of gameplay requirements, they also indicated a knowledge gap in our understanding of these requirements, and in turn called for further research in the area. Moreover, the 2017 paper of Scacchi indicates that knowledge of one type of games (e.g. mobile phone games) “does not subsume, contain or provide the gameplay experience in other games” and that “being skilled in development of for one type of games, e.g. MMORPG, does not imply ability or competence in developing software for another type of game, e.g. a continuous play twitch” (Scacchi, 2017), p. 113). Understanding therefore the unique context of engineering gameplay requirements and the ways in which they are conceptualized is important to advance our knowledge in the field. The present research directly responds to these calls and extends the conversation on gameplay requirements to the realm of MMORPGs. To the best of our knowledge no empirical RE study has been published on the particular context of MMORPGs.

In order to get an in-depth understanding of how gameplay RE happens in real life, we executed an exploratory study (Yin, 2008) by involving practitioners that are professionally engaged in developing MMORPGs. Rather than investigating discrete aspects of the gameplay RE practice, our study sought a broader perspective on gameplay requirements and on the range of possible ways in which the practitioners reason about them and approach them in their projects. Sixteen practitioners from three leading MMORPG-developing and publishing companies have been interviewed and their gameplay requirements documents were reviewed in one-on-one walkthroughs with the researcher. Using qualitative data analysis techniques from the Constructive Grounded Theory (Charmaz, 2007), the study revealed how gameplay requirements have been reasoned about, from the perspectives of those working in the field. The key result of the study is a contextualized description of the practitioners views on gameplay requirements and on the ways they are coped with. The findings are compared with those from published relevant studies in the fields of human-computer interaction, social behavior, and psychology, and some implications are drawn for research and practice.

We make the note that first results of the study have been presented at the 2014 international conference on Requirements Engineering (RE'14 (Daneva, 2014)). The present paper extends the 10-pages conference publication by providing new results in the form of new concepts that came out of our detailed data analysis, plus richer and deeper description of those concepts already mentioned in the 2014 paper. Next, in this paper we provide an in-depth analysis of related work by including empirical studies on RE for games, and systematic literature reviews published until May 15, 2017, which we use to evaluate our results in light of published findings of other authors. Third, we provide a detailed methodologically-grounded description of the choices behind our research design. Fourth, we extended discussion in depth and breadth (e.g. by including aspects such as theory-building, realism of generalizability claims among other topics).

As already indicated in Daneva (2014), our research on gameplay requirements for MMORPG extends previous empirical RE and empirical SE on game systems in several ways. First, we add to the body of empirical RE research on game systems by providing insights into one particular type of requirements (gameplay) in one specific context (MMORPGs) that so far is unaddressed in empirical RE literature, and from the perspective of practitioners working in that context. In this sense, our case study helps closing an existing gap of knowledge (Callele, 2005,2009; Alves, 2007;).

Second, the paper advances our understanding of gameplay requirements as functional requirements, which complements the earlier publications on games requirements (Callele, 2005) in which gameplay requirements were treated as quality requirements (or as also called ‘non-functional requirements’). As we will see, the findings of our study could be possibly generalizable to other similar but different settings.

The paper also provides a direct response to the call of the empirical SE community for more empirical research on using SE processes in specific contexts (Sjøberg et al., 2007). The MMORPG requirements as artefacts and the RE processes for MMORPGs have largely been overlooked by empirical SE researchers, as it is evident from the systematic review of Ampatzoglou and Stamelos (2010). In the SE field, research about games has so far been constructionist in nature (e.g. the GAS series of workshop (Bell et al., 2012) at the ICSE conference), meaning that considerable efforts

were devoted to construct methods that address a variety of technical, network and infrastructure aspects of MMORPG game development, e.g. by taking the perspectives of artificial intelligence, multi-agent systems, autonomic operations, and peer-to-peer protocols on games. Collecting evidence from an exploratory study on MMORPG gameplay, therefore, complements these perspective by adding a practitioners' view on the processes and the technologies used to "engineer fun" (Fullerton, 2008).

Next, our study contributes to the conversation recently initiated in both SE and RE communities on the contextual differences between game development projects and development projects delivering 'traditional' systems (e.g. business information systems). The 2017 publication of Scacchi (2017) and the 2017 empirical study of Valente et al. (2017), as well as the 2014 study of Murphy-Hill, Zimmermann and Nagappan (Murphy-Hill et al., 2014) on the differences between game development projects and 'traditional' software development projects, signal shortage of empirical evidence on processes and practices from game project contexts. These authors argue that the lack of knowledge of the differences between game development contexts and 'traditional' system contexts poses at least two risks: first, by being unaware of the peculiarities of game development, we miss on an important opportunity to deliver methods and tools that are directly applicable to game development; and second, by being unaware of the contextual similarities between game development and 'traditional' development projects, we miss on an important opportunity to leverage RE techniques that have already been empirically evaluated (i.e. true and tested ones), in the game development context. Requirements professionals working in the game sector would possibly benefit from using those techniques. This paper contributes an empirical foundation on which to understand how RE for games – in particular MMORPGs, is different from 'traditional' RE as described in well-known textbooks (e.g. Lauesen, 2002). Specifically, our exploration indicates that MMORPG gameplay requirements are a multilayered concept and while some layers could possibly be coped with by adapting or adopting 'true and tested' RE techniques, others demand re-evaluation of what we know about RE by borrowing theories from other disciplines.

The paper is structured as follows. Section 2 provides background on MMORPGs as online businesses and as systems. It also includes related work on play-centric game design and RE, and on conceptualization of gameplay. Section 3 describes our research process and its execution. Section 4 presents the results and Section 5 discusses them from the perspective of previously published literature sources. Sections 6 and 7 extend the discussion of the results by evaluating validity threats and implications for RE research, practice, and education respectively.

2. Background and related works

2.1. MMORPG as online businesses and as online systems

This section informs less familiar readers on MMORPGs as online businesses and as systems. MMORPG sites are powerhouses of e-commerce, forming part of the entertainment services industry. Similarly to other games, MMORPG usually use two business models, subscription-based and 'freemium'. The first generates profit out of a growing long-term subscriber base and, hence, player retention is key to MMORPG producing companies' financial health. In this model, subscribers buy the game (client) software for, e.g. 50 Euro, and pay a monthly fee of e.g. 15 Euro, to access the game online (server). In contrast, the freemium e-business model (Anderson, 2009) assumes the presence of a huge base of free clients and a percentage of it to turn into paid clients at some point of the process of playing the game. Usually, in this model the producer gives away the game's core functionality for free, while

offering upgrades to add certain features for a small fee. Converting free players to paid ones at a consistent rate is strategic to such a producer. In both business models, to thrill players and keep them playing, MMORPG companies vigorously embrace innovation in both signal processing and game design, and commit to deliver new, high quality game contents (e.g. extensions, patches) on regular basis. Gameplay is critical to MMORPG-companies' client retention as it's in fact what keeps players stay subscribed to play longer (Debeauvais et al., 2011; Riekk, 2015).

As a system, a MMORPG provides players with functionality comprising an extensive range of game abilities, including character design options, weapons, tools, statistics, all of which grow in number and intensity proportionate with player's progress through the game. In a MMORPG, players live through their characters (the so-called 'avatars' which are virtual bodies "created by users to project their identity and actions into the online world" (Ducheneaut et al., 2009)), and compete with other players' avatars and non-player characters (NPCs), e.g. dwarfs, zombies, ancient stone keepers. While playing, players can switch between two perspectives: the first person and the third person perspective. Also, based on whether players fight against other players or NPCs, fighting happens respectively on Player versus Environment (PvE) or Player versus Player (PvP) servers (Quax et al., 2014). In both, players use weapons and spells to deal damage to the target and upon success they are awarded experience points that are critical to moving from one game level to another. For a player, creating an avatar implies choosing a name, gender, race, social class, and facial features. We make the note that the MMORPG avatar is more than a virtual projection of the player: unlike getting a predefined character as in games such as Angry Birds, the MMORPG player develops it to his/her own taste within the game world's rules (Klevjer, 2012; Bainbridge, 2010). Avatars necessarily join one of two conflicting factions – good and evil, that build up the storyline for each player. Choosing a faction implies each player enters the game universe through this faction's own home city, and that they design their avatars based on the options available to the faction exclusively. A player's avatar may choose a profession, e.g. the World of Warcraft, a popular MMORPG, offers eleven professions – alchemist, enchanter, engineer, cook, fisherman, herbalist, scribe, jeweler-crafters, leatherworker, miner (blacksmith), and tailor, which avatars gradually master from beginner's to highly advanced level. These social occupations are critical to the way the players plays the game because they create the ground for cooperating and entering economic transactions with other players (Bainbridge, 2010). As a player's avatar advances throughout the game and its abilities gradually grow, the number of potential approaches to in-game situations grows too, e.g. quests, fighting, crafting and teaming-up. Players' progress is structured in terms of levels organized according to a common template for avatar development (Klevjer, 2012), that could be stereotyped as follows:

- (i) players start the game by creating level 1 avatars first;
- (ii) the avatar enters the online world with some predefined limited set of abilities and equipment;
- (iii) the avatar necessarily is confronted with missions to accomplish;
- (iv) successfully accomplishing its missions earns the avatar experience points which translate in acquiring more powerful abilities and sophisticated equipment;
- (v) the difficulty of the missions grow to a point that the avatar can no longer accomplish a mission alone and necessarily joins a team of other players' avatars to coordinate and execute mission's activities;
- (vi) the missions get too resource and time demanding and the challenges they pose can be no longer be dealt with by teams built up on an ad-hoc basis, which necessarily leads

to forming social structures of a more permanent nature, in order to keep playing.

This ‘progress template’ explicitly indicates that preparing for fights and fighting gets increasingly more social and group-based, as levels advance. Once a player passes the end level of a MMORPG, the only way to continue playing endlessly is by joining a group. Also, the template implies that for players to move ahead on the levels, they need to acquire deep knowledge not only of their own avatars’ functions but also of the functionality of other players’ avatars. Plus, players should be able to use this knowledge to anticipate the moves of other players’ avatars and NPCs. E.g. the study of Williams et al. (2014) indicates that in the Cataclysm expansion of the World of Warcraft, there are 100 unique NPCs (i.e. monsters) to encounter within this expansion’s five levels, each with a host of abilities to be understood.

2.2. Play-centric game design and the role of requirements engineering in it

MMORPG companies usually use play-centric approaches to game development that span from idea conceptualization to completion (Fullerton, 2008). Central to them is the goal to achieve playable and satisfying game experience; this means continually keeping the player experience as the driver in making design choices through the game development process, and testing the gameplay that these choices create with target players at each development stage (Fullerton, 2008). In a play-centric development context, RE is an agile process as it’s not limited to the early stages of game conceptualization; instead, it accompanies the play-testing and the prototyping activities as assumptions about player requirements might render unrealistic or new requirements may be discovered. Also, RE is an on-going process because it goes well beyond MMORPG launch and active play as companies keep collecting players’ feedback via online communities, which is an important way to get requirements for upgrades and new releases.

Prior research on RE for games did acknowledge the play-centric nature of requirements processes (e.g. LeBlanc, 2006; Zagal and Bruckman, 2008) and the prominent role that enjoyability, fun, emotions and gameplay take in game development (Zagal and Bruckman, 2008; Bjoerk and Holopainen, 2005; Deaker, 2012; Kasurinen, 2014). However, in the field of RE, very little research focused on gameplay specifically. The 2010 systematic review on SE for computer games (Ampatzoglou and Stamelos, 2010) found 37 empirical studies on RE for games, but only two (Callele, 2005; Alves, 2007) included gameplay requirements as a sub-topic in their analyses, in the context of video games and mobile games development, respectively, e.g. the study of Callele (2005) presents some techniques practitioners used for gameplay prototyping in video game design. Both the studies of Alves (2007) and Callele (2005) point to these requirements as a particularly challenging area for which more research is needed. However, none of these studies elaborated on gameplay requirements in detail, nor dealt with gameplay requirements exclusively nor considered gameplay in MMORPG context. Next, the 2013 systematic review of Orita Almeida and Corrêa da Silva (2013) on game design methods found 19 studies that deal with game-design-centric techniques. The authors indicate a common trend in these studies towards building shared vocabulary about game design elements and developing visual modelling languages. In these 19 studies, the concept of gameplay is referred to as something inherent to games, however its meaning is not defined or explained. There is a tacit assumption that the readers of the respective studies know its meaning and its interpretation in context.

As part of preparing this paper, we searched the Scopus digital library for (1) relevant RE studies on games published after

2010 – the publication year of (Ampatzoglou and Stamelos, 2010), and (2) relevant systematic reviews on MMORPGs. The focus of first search was to achieve depth, while the focus of the second search was to achieve breadth. The first search was done on May 12, 2017 and yielded 13 publications (Pasquale, 2013; Cooper, 2014; Kasurinen, 2014; Guiard et al., 2014; Quax et al., 2014; Östblad et al., 2014; Mueller et al., 2014; Callele et al., 2015; Paschali et al., 2014; Scacchi, 2017; Djaouti et al., 2008; Teruel et al., 2016; Valente et al., 2017) that include the topic of RE for games, in addition to Callele (2005) and Alves (2007) which were included in the 2010 systematic review of Ampatzoglou and Stamelos (2010). Among those studies, Pasquale (2013) and Mueller et al. (2014) are concerned with console-based games (e.g. Wii), (Cooper, 2014) – with serious games, (Kasurinen, 2014) – with the role of RE in (non-MMORPG) game development including various platforms, (Guiard et al., 2014) – with interactive computer games for children with autistic spectrum disorder, and (Östblad et al., 2014) – with audio-based adventure games for blind or sighted players. Table 1 compares and contrasts the publications that dealt explicitly with game and gameplay requirements, although not all of these take a RE perspective.

In fact, three studies Pasquale (2013), Cooper (2014) and Kasurinen (2014) are published in RE venues and five elsewhere. Among these studies, Kasurinen (2014) is the only one in which gameplay requirements and related concepts such as game mechanics, prototyping of requirements and balancing games’ rules are discussed explicitly.

We note that Table 1 includes only one study (Quax et al., 2014) dedicated on MMORPGs exclusively. Two studies are focused on computer games (Paschali et al., 2014) and collaborative games (Teruel et al., 2016) in general, and as part of their research they pay attention to MMORPG as a particular sub-area. The goal of the study of Quax et al. (2014) is to classify the games in the MMORPG marketplace and therefore these authors defined the term ‘gameplay’ strictly for the purpose of MMORPG classification. As per this study (Quax et al., 2014), gameplay is the way a player plays in terms of whether he/she fights against NPCs or against other players, meaning that gameplay is something that happens either in a PvE server or in a PvP server. Next, Paschali et al. (2014) present an exploratory survey with game players of a wide range of computer games including MMORPG, on those non-functional requirements (such as solidness of the character, quality of the graphics among others) that impact players’ enjoyment. While there is no definition of the term ‘gameplay’ in this study, the authors hint that it is the player-game interaction that helps realize enjoyment. Last, the study of Teruel et al. (2016) focused on the space game awareness as a non-functional requirement for collaborative games – including MMORPG, and proposed and evaluated a method for engineering those requirements. Gameplay requirements are not explicitly defined.

Table 1 also indicates that gameplay requirements have been the explicit focus of three papers (Kasurinen, 2014; Holopainen, 2011; Djaouti et al., 2008) that however explored the domain of video games (Holopainen, 2011; Djaouti et al., 2008) and non-MMORPG (Kasurinen, 2014). Djaouti et al. (2008) defined gameplay as “the association of “game rules” stating a goal to reach, with “play rules”, defining means and constraints to reach this goal”. These authors proposed the Gameplay Bricks approach to gameplay specification, this means using “recurrent diagrams” to express the rules of video games. The authors use their approach to come up with typology of video games while requirements elicitation (or RE in general) was outside their specific scope. Next, the PhD thesis of Holopainen (2011) defined gameplay as “the interaction between the game rules, challenges, elements, and players”. This author argued that a game is defined by its gameplay and focused its exploratory efforts on the structures of gameplay as de-

Table 2
Related systematic reviews on MMORPGs.

Systematic review	Topic	Theoretical perspective	What did the review examine?	What does the review say about gameplay in MMORPGs?
Caroux et al. (2015)	Concepts of player-video game interactions in entertainment situations	Human-computer Interaction (HCI)	Player's aspects (enjoyment and engagements) and video game aspects (input/output techniques, multiplayer aspects and game contents).	Gameplay is a distinctive characteristic of player –video game interaction compared to other HCIs. Gameplay is a game aspect related to contents. A game session is a unit of contents in a video game.
Mekler et al. (2014)	Enjoyment of digital entertaining games	Players experience (PX)	Quantitative measures of game enjoyment and how these relate to players experience components (flow, pleasure and immersion)	Gameplay is what creates flow and stickiness
Lopez et al. (2013)	Models of theoretical elements of leadership in serious games	Leadership theory, social learning theory	Pedagogical and technical features in 'serious games' (incl. serious MMORPGs)	Gameplay is the conduit to interactivity and supports a pedagogical goal
Sublette and Mullan (2012)	MMORPGs and mental health addiction	Psychosocial effects of MMORPGs	Physical and psychosocial variables correlating with playing practices of MMORPG players	Gameplay as addictive mechanism
Warmelink and Siitonen (2011)	Players communities in MMORPGs	Sociological perspectives (community, guild, group)	Concepts to characterize players communities and the operationalizations of those concepts	Gameplay is implicitly related to social structuring (i.e. how players define roles specific to communities).
Boyle et al. (2012)	Engagement in digital entertaining games	Human behavior	Aspects of players' engagement in games, incl. the physiological concomitants of players' experiences, motivation for playing, game usage and time spent playing games and the impact of playing on life satisfaction	Gameplay is a vehicle for players' engagement in a game

sign material. The author proposes and evaluates a pattern-based approach to both analyzing existing games and aiding in designing new games. However, Holopainen's work treats gameplay from game design perspective, and not from RE perspective (which is the focus of this paper). Last, the exploratory study of Kasurinen (2014) set out to uncover the RE practices used in Scandinavian game development companies. These authors identified that while practitioners do not follow any rigid RE process model and use no particular RE terminology, they do use RE practices for the purpose of requirements identification and analysis. Regarding gameplay in particular, the study indicates that companies use “basic gameplay elements” to structure the contents of the games under development.

2.3. Gameplay in MMORPG

Furthermore, the second search for related work was done on May 15, 2017 and yielded six systematic reviews addressing gameplay as part of research on MMORPGs (see Table 2). These reviews are from the fields of human-computer interaction (Caroux et al., 2015), psychology (Sublette and Mullan, 2012; Mekler et al., 2014; Boyle et al., 2012), business leadership development (Lopez et al., 2013), and sociology (Warmelink and Siitonen, 2011).

Table 2 indicates that the concept of gameplay lends itself to investigation from a broad number of theoretical perspectives beyond the SE discipline. Below, we provide some examples based on the references found in these reviews, that illustrate how gameplay is conceptualized in other disciplines. In the fields of human-computer interaction (Caroux et al., 2015) and social psychology, researchers investigated usability (Cornett, 2004), playability (Gonzales, 2013), game engagement (Nardi, 2010; Brockmyer, 2009), gameplay experience (Pereira and Roque, 2013) and aesthetics aspects (Bergstroem, 2010) of video games and MMORPGs, with a particular focus on gameplay. In the field of game design, gameplay was conceptualized by employing layered models, e.g. LeBlanc's Mechanistic-Dynamics-Aesthetics (LeBlanc, 2006), or gameplay structures that mean the players' interactions in the

multi-play happening in a game (Bjoerk and Holopainen, 2005). Approaches also have been developed to capture and document gameplay design patterns, e.g. the game ontology project (Zagal and Bruckman, 2008) that employs a hierarchical structure to describe gameplay elements from players' perspective, and the 400 rules project (Falstein, 2004) that stores and shares knowledge in 'rules' format, from professional game designers. These studies on gameplay can be useful for understanding games and games mechanics but they do not directly address gameplay requirements and how they are handled. However, following Callele (2005), we think that even if these studies do not address RE, the techniques they proposed could possibly form an interesting extension to the RE approaches already available to practitioners.

3. The exploratory study research plan

3.1. Research goal and research questions

The goal of this study is to understand how requirements elicitation and discovery happens in MMORPG projects. To this end, we set out to answer the following research questions (RQ):

RQ1: What do gameplay requirements mean to practitioners involved in MMORPG development projects?

RQ2: What MMORPG project artefacts are concerned with gameplay requirements? and

RQ3: How do practitioners manage the process of gameplay RE?

RQ1 attempts to reveal the concepts that practitioners in game software development use when talking about gameplay requirements for the games they work on. We wanted to know the variety of angles on gameplay requirements that may exist in the field and that, in turn, may possibly influence how practitioners approach these requirements in their working processes. We assumed the presence of multiple possible perspectives on gameplay requirements because RE textbooks (e.g. Lauesen, 2002; Kotonya and Sommerville, 1989) consistently suggest that any large-scale

project necessarily includes multiple viewpoints and expertise areas and that being aware of the commonalities and the differences among those is key to the RE process success.

Next, RQ2 and RQ3 reflect the current understanding of the RE community that the study of any RE phenomenon could be approached with two types of principles in mind: artefact orientation principles and activity orientation principles (Méndez Fernández and Penzenstadler, 2015). When following the principles of artefact orientation (RQ2), we start by defining RE artefacts, their contents, and their dependencies rather than thinking of the way of creating the artefacts. In contrast to this, process orientation principles (RQ3) put the process thinking first and are based on the idea of an organization's adoption of an ordered set of activities and methods, each defining procedures and techniques for a particular purpose (Nuseibeh and Easterbrook, 2000), from which project teams can select the appropriate one to design their project-specific RE process.

Our research design followed Yin's guidelines for exploratory studies (Yin, 2008). As Bensabat (1987) suggest, a case study is a particularly suitable research method to situations in which (1) researchers study socially constructed processes in systems development projects and seek to achieve as good as possible grasp of reality, and (2) researchers want to answer 'how' and 'why' questions and comprehend the nature and the complexity of the processes of interest. Drawing upon this, we expected the case study research method would let us earn a rich and contextualized understanding of the practitioners' meanings behind the concepts of 'gameplay' and 'gameplay requirements' and the ways in which practitioners approach the engineering tasks essential to these requirements.

3.2. Study design, participating companies and informants

We designed a research process including semi-structured open-end in-depth interviews with 16 practitioners from three producers that are highly visible in the MMORPG market. Because of confidentiality agreements, we would not be able to provide any information on the country location of these producers, nor on their organizations' size. However, the MMORPG companies shared the following common characteristics:

- (i) they all use the subscription-based business model and player retention is their key strategic objective;
- (ii) all three games included in the study are known to be profitable ventures;
- (iii) each company had at least one million subscribers,
- (iv) they all follow play-centric approaches (Fullerton, 2008) to MMORPG design;
- (v) all three games have a common goal, namely to increase the fitness of the player's character, which is endless in nature and not bound to a particular game level (because keeping players playing is in essence player retention);
- (vi) all three were real-time strategy games, meaning that playing requires constant attention to actions in the game in real time;
- (vii) the companies all built in-game and out-of-the-game communities of players and have the continuous improvement of the players' social collaboration as an underlying motivation for MMORPG development projects; they produce packs, extensions, enhancements, on annual basis. Because of confidentiality agreements, we refer to the companies as to TechGame, ArtGame and YouGame.

Table 3 summarizes the information about the participating companies and practitioners. Their job titles and project roles were diverse, including various artistic specialists, play-testing managers, level designer, community development manager, playability analyst, gameplay analyst, helpdesk analyst, web developer, senior software engineers (e.g. on client and server sides).

Our study was carried out in four steps: (1) Prepare an interview guide following the guidelines in King and Horrock (2010); (2) Do a pilot interview with one practitioner to check the applicability of the guide to real-life context; (3) Carry out interviews with practitioners according to the finalized questionnaire; (4) Follow-up with those participants that possess deeper knowledge or a specific perspective. We note that after the first interview (the pilot), there were no substantial changes in the interview guide and for this reason, we included the pilot interview as a legitimate one in our data analysis (as suggested by Alves, 2007). The interview guide is receivable from the author upon requests of interested readers.

The time each interview took was between 90 and 100 minutes. Six interviews were face-to-face, and ten - over a conference bridge with shared presentation screen facility used for walking through a gameplay document that the respective interviewee presented to the researcher. As strict confidentiality agreements were discussed before the interviewees gave their consent to participate, each interviewee was informed by the author on the research process and on how data will be handled. The author provided each participant with the list of interview questions two weeks before the interview date. Each interviewee was also asked to bring a gameplay document to be used for illustration purposes during the interview, so that the researcher makes sure she understands what the practitioner means when using a specific gameplay-requirements-related concept. Each interviewee came prepared to the interview, which was key to the effective use of the interview time slots.

3.3. Selecting and characterizing the informants

We used two selection criteria for practitioners' participation. The participants should:

- (1) have exposure to the development context of MMORPGs; and
- (2) be involved in RE tasks as part of their professional responsibility in the company.

'Involved' meant that at least 30% of the work time that a practitioner logs in a project goes to work tasks or activities related to 'gameplay'. As Table 3 shows, we had two practitioners (out of 16) that spent 30% of their work time on gameplay requirements tasks, three that said gameplay requirements are the very core of their jobs and they spend 100% of their time on them, six practitioners who spent 50% of their work time on gameplay requirements and four - 75%. Tasks can broadly range from writing gameplay requirements, doing analysis of gameplay requirements by using in-game statistics, to attending review meetings, building prototypes and providing feedback.

The practitioners were chosen because their jobs pertained to this study's research context. Their experience included at least three projects concerning the MMORPG produced by their companies. All practitioners were avid players of their respective company's MMORPG. Their playing experience was at least 6 years. We note that the hiring policies in all three companies explicitly stated preferences for job candidates who had playing experience. In fact, all 16 practitioners had met this condition when they applied for their positions in their respective companies. Last but not least, the practitioners were excited about this exploratory study, they felt enthusiastic about winning a researcher's interest in their professional activity. They all thought, researchers could help industry by promoting careers in game design, to the students they teach.

To get access to the practitioners, the author used as mediator a member of the International Game Development Association (IGDA, www.igda.org). The IGDA member chose to remain anonymous and not participate as a co-author of this paper. The mediator was a former colleague of the author and was instrumental to

Table 3
The case study companies and participants.

Company name	Company Details	Participant ID	Participant Designation	Estimated work time spent on gameplay requirements tasks	Years of playing experience in the respective MMORPG
TechGame	North-America-based game developer and publisher with a Player Support Centre in Europe and an international player base	P1	Level Designer	75%	8
		P2	Playability Analyst (P2)	30%	7
		P3	Senior Software Engineer – Server Systems (P3)	30%	12
		P4	Senior Software Engineer – Client Systems	50%	9
		P5	3D In-Game Artist – Cinematic Animation	75%	10
ArtGame	North-America-based game developer and publisher with a Player Support Centres in Europe and Asia, and an international player base	P6	Environment Texture Artist	50%	8
		P7	Character Artist	50%	9
		P8	Project Lead	50%	7
		P9	Play-testing Manager	100%	8
		P10	Helpdesk Analyst, In-game Issues	75%	7
YouGame	East-Asia-based game developer and publisher with operations in Europe and an international player base	P11	Web Front-end Engineer	75%	7
		P12	Community Development Manager	100%	7
		P13	Visual Effect Artist	50%	6
		P14	Project Lead	50%	8
		P15	Gameplay Analyst	100%	8
		P16	Play-Testing Manager	100%	7

hook up the author with relevant parties in the three game producing companies. Thanks to the mediator, the author also made a visit at the European branch of one of the companies and executed the six face-to-face interviews. The mediator suggested a broad range of possible contacts however the author used purposive sampling (King and Horrock, 2010) to select the 16 interviewees so that they represent a broad variety of viewpoints, roles and contributions to gameplay RE for MMORPG.

3.4. Our data collection

We used the method of ‘focused semi-structured’ interview (King and Horrock, 2010) for data collection. ‘Focused’ means that (1) gameplay, (2) gameplay requirements and (3) managing gameplay RE were the focus of all conversations with the 16 practitioners. ‘Semi-structured’ means that a list of interview questions is prepared in advance, however, during the interview the researcher has the right to ask new questions (e.g. to explore a specific perspective on an issue and probe for details) or to drop existing questions based on his/her judgment of the opportunities to extract rich contextual information in the conversation with the interviewee. Choosing the semi-structured interview method ensured our data collection was shaped by both the participant’s own understandings of gameplay and gameplay requirements as well as the researcher’s interests. Also, it allowed us to take unexpected routes as opportunity arose. We note that interview-based exploratory studies usually are intended to promote self-disclosure and that is what we were after in this work. As in King and Horrock (2010), interview studies are not used to provide statistically generalizable results applicable to all professionals with profiles similar to those of the practitioners in a specific study. The intention of the exploratory case study is not to infer, but to understand and not to generalize, but to determine a possible range of views. There-

fore, in this study we will adopt, based on the recommendations in King and Horrock (2010), the criterion of transferability as a useful measure of validity. Transferability asks for whether the results are presented in a way that allows other researchers and practitioners to evaluate if the findings apply to their contexts.

3.5. Our data analysis

The focus of our data analysis was to identify and explicate (1) the working practices of the practitioners associated with gameplay, and (2) the elements of context that contributed to generate the emergent forces. Our data analysis was inspired by the guidelines in Charmaz’ constructive grounded theory (GT) method (Charmaz, 2007), a qualitative approach commonly deployed in empirical studies in social sciences to construct general propositions (called a “theory” in this approach) from verbal data. The GT analysis is exploratory in nature and suitable to research settings where the researcher does not want to use pre-conceived ideas, and instead is driven by the desire to capture all facets of the collected data and to allow the propositions to emerge from the data. For the purpose of our data analysis, we chose Charmaz’ set of guidelines over other GT methodological sources (e.g. those described in Evans, 2013), because of its positioning of the researcher in relation to the participants and its rendering of participants’ experiences, which matched our situation: We wanted to focus on interpretative understanding of participants’ meanings and Charmaz’ constructive GT approach lends itself to this use, as it helps a researcher incorporate multiple participants’ voices and focus on each participant’s experience “in its fullness” (Charmaz, 2003). In contrast to other GT approaches (Breckenridge et al., 2012; Mills et al., 2006; Evans, 2013) that aim at a conceptual understanding of social behavior, constructive GT helps a researcher develop interpretative understanding of practitioners’ meanings and this is what

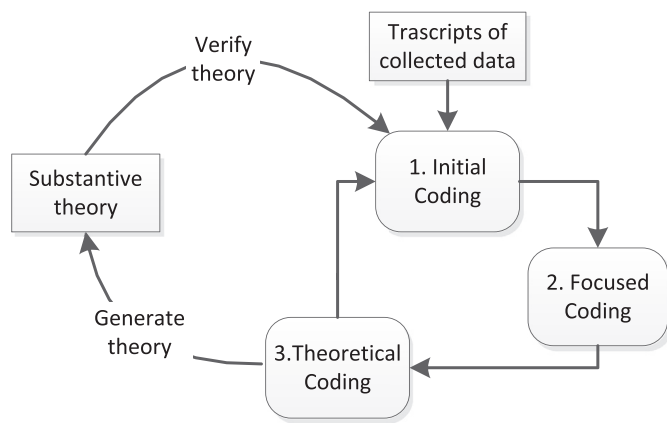


Fig. 1. The coding process in our study (according to Charmaz' guidelines (Charmaz, 2007)).

we were after in this study. Next, constructive GT recommends researchers privilege the participant's main concern over seeking accuracy from researcher's professional perspective. (For more information about the philosophical underpinnings of the constructive GT and how it is different from other approaches, we suggest interested readers to look at the methodological papers of Breckenridge et al. (2012), Mills et al. (2006) and Evans (2013).

The essence of the GT guidelines is in making analytic sense of the interview data by means of coding and constant comparison of pieces of data that were collected in the case study. Regarding coding, the constructive GT approach suggests the process of initial line-by-line coding, focused coding and theoretical coding (see Fig. 1).

Initial coding means the first level of coding in GT analysis, in which the researcher labels and assigns units of meaning to pieces of text that describe perceptions on a phenomenon, an event, or an action. Focused coding is the process in which the researcher identifies preliminary themes and concepts emerging from the data. The researcher focuses on the most significant and frequently occurring codes (Charmaz, 2007). Theoretical coding is the activity of the researcher's merging concepts into thematic categories. The resulting grounded theory is constructed from analysis of the interrelationships among the themes.

As recommended in GT methodology (Charmaz, 2007; Breckenridge et al., 2012; Mills et al., 2006; Evans, 2013), all stages incorporated signature GT processes of constant comparison, whereby data are continually compared and contrasted at each level of analysis (see Fig. 2). Constant comparison means that the data from an interview is constantly compared to the data already collected from previously held interviews (Fig. 1). The data analysis process deployed in this study, leveraged the author's prior experience in empirical studies using GT (e.g. Daneva, 2013; Daneva et al., 2014). As in these previous studies, our data analysis in this study started with reading interview texts and labeling a portion of the text – a phrase or a paragraph, with a 'coding' word. The 'codes' mark the meaning of the respective portion of the interview text to a specific research question. A code can be a concept (e.g. 'look and feel', 'playability', 'player experience') or an activity (e.g. 'balancing the gameplay', 'play-testing', 'paper-prototyping'). All pieces of text that relate to the same code are then clustered in order to analyze it in a consistent and systematic way. While coding, the language of the participants was used whenever possible, so that the results of the qualitative data analysis remain as close as possible to the primary data (the practitioners' original perceptions), as per the guidelines of Charmaz (2007) and Daneva et al. (2014). The results of the data analysis and the discussion on them are in Sections 4 and 5.

4. Findings

In this section, the codes – concepts and activities, are marked by using an underlined italicized font in their first appearance, e. g. play-testing is marked as *play-testing*. As it is usual in qualitative studies, we supplement the findings with interviewees' quotations. These are given in *italic*.

4.1. The meanings of gameplay and gameplay requirements

From RE perspective, our analysis suggests gameplay requirements refers to a group of requirements – and their interactions, that when satisfied together, deliver player's experience that matches or exceeds the players' expectation. In our participants' views, the interactions between the requirements in the group heavily contribute to the overall effect on player's experience. While our participants agreed on this understanding of gameplay requirements, they differed regarding how they think of the term 'gameplay' itself. Our findings suggest no consensus on what gameplay is. The interviewees hinted to three ways to conceptualize gameplay: as a process, as an artefact, and as a relationship between game creators and players.

4.1.1. Gameplay as a process

Five practitioners (P3, P6, P9, P10, P16) thought of *gameplay as a process* that a player has to go through in order to achieve the end of a level. In essence, they made a clear distinction between two types of gameplay-as-a-process (as depicted in Fig. 3):

- (1) '*within level*' *gameplay*, which is bounded to a level in the game (e.g. level 2 in a game of 99 levels); and
- (2) '*end level*' *gameplay*, which is the one that happens once players exhaust all levels and start consuming the so-called 'end-level game content' (e.g. once one is at level 99).

In the experience of the five practitioners, gameplay as a process – both within a level, and at end level, was deemed specific to the *role* a player takes. E.g., in the game that is produced by the case study company ArtGame, a *healer* is the role being tasked with keeping other players alive, and a *shooter* is the role being tasked with eliminating enemies. At each level, the gameplay process is perceived as a series of role-specific tasks arranged in three groups: preparing the avatar for coping with the in-level challenges, execution, and wrap-up. For example, an avatar that plays the role of healer receives always tasks that support the goal that each and every one of his fellow raid members survive until the respective raid's monster is dead or until the healer exhausts his resources. The healer's tasks will start only after some raid members suffer damage, and then the healer will use its healing spells to heal the damaged fellow members. The task execution is heuristic in nature due to a few factors: (1) different healing spells are suited for different situations. Each healing spell is traded off against alternative spells applicable to the situation, based on its research cost and cast time (e.g. high resources cost and low cast time versus low resource cost and high cast time). (2) the damage the healer's raid takes is not always predictable; fellow player's error may often cause random bursts of damage which the healer would have to heal. (3) various abilities of the respective monster can target healers, shifting the balance of his/her task execution. The healer would need to improvise. In other words, in order to achieve the goal of his role, the healer would need to make good use of his/her spells and resources at the correct time.

The tasks may vary broadly in terms of contents, complexity, and player's time and skill demands (as shown on Fig. 2). However, once the end level of the MMORPG is achieved, the gameplay process' goal is endless and focused solely on equipping the avatar as

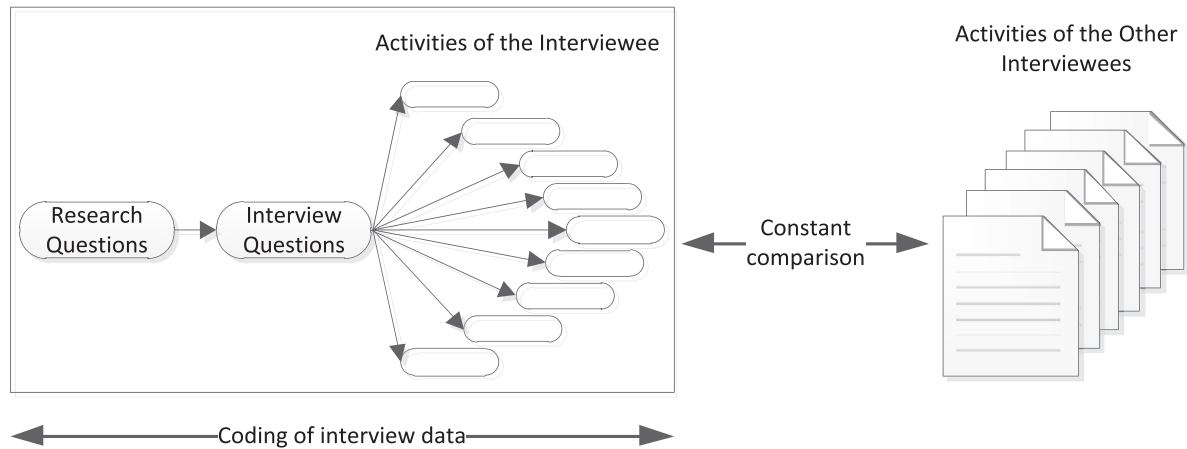


Fig. 2. The constant comparison of pieces of data of one interview with pieces of data from other interviews: a conceptual model.

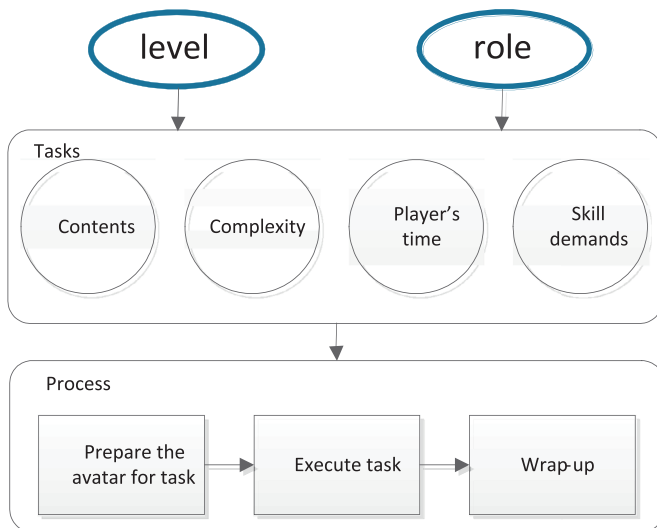


Fig. 3. Thinking of gameplay as a process.

well as possible. e.g. the endgame experience is focused on organizing for large-scale heavily-coordinated attacks among up to 60 players, where players are expected to commit significant amount of time (up to 18 hours weekly) in configuring the avatar in a way that makes it 'fully prepared' to cope with the challenges in the run. 'Fully prepared' does not only mean accumulating weapons, magic spells and food, but also adding layers of protection against the various dangerous aspects of encountering the level's NPC (i.e. a monster).

4.1.2. Gameplay as an artefact

Next, three practitioners (P1, P8, P13) thought of *gameplay as an artefact*. To them, this is a set of rules, penalties and awards within a game level, all intended to assert game structure, to control or to influence the players' in-game behavior. As an artefact, gameplay describes the admissible actions, definitions and constraints that apply to the MMORPG world. These should be consistent and non-conflicting, and can apply to avatars, NPCs, weapons, spells, and anything else in the game. For example, *mode-of-playing rules* are specific to the server type that players may choose to

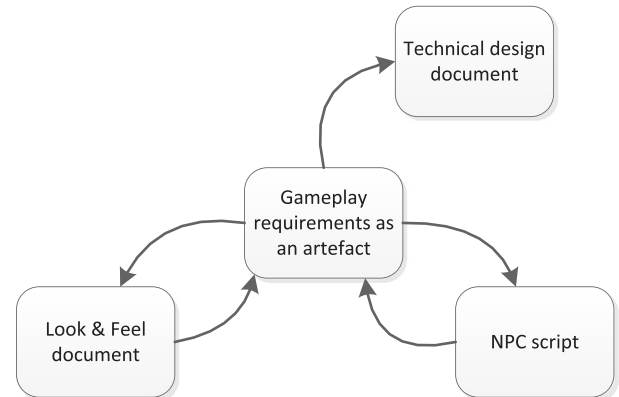


Fig. 4. Linkages between the gameplay requirements as an artefact and other documents.

use. MMORPGs usually distinguish among three server types that determine three modes of playing: PvE, PvP and role-playing (RP) servers. E.g. a very important rule for the PvE mode is that players cannot kill other players by default, unlike in the mode of the PvP servers. Moreover, in RP mode, players are supposed to take on the role of a character and act it out in-game through emotes. Based on whether a player commits to follow the RP mode rules or not, its status is *in-character* (which means the player commits to posting emotes), or *out-of-character*, respectively.

An interesting dimension of gameplay as an artefact is its linkage to two non-technical documents and one technical (see Fig. 4). First, 'the Look and Feel' document of the game:

"The game designer comes up with the look and feel, a page long description, and then the first cut of the gameplay tells you what it is like to be in the game; you try out the gameplay to get into the looks and feel, and you've got to tell the game designer if the gameplay you experiences is indeed the look and feel he wants in the game." (P1, TechGame).

In the views of our participants, while the Look and Feel document is artistic and expresses a vision for the game, it's the gameplay that is the first "materialization" of this vision. Practitioners thought of it as "the first prototype of the look and feel":

“Think of gameplay as the first prototype of your look and feel; you’ve got to start from somewhere in your game development and here is your starting point.” (P8, ArtGame)

“No team fellow can contribute to the game unless he/she is aware of the look & feel. Achieving this look& feel is what drives your decision-making.” (P13, YouGame).

Second, the so-called “NPC script”. Each of the games in the three games companies, participating in this study, included between 30 000 and 40 000 NPCs. Many of them talk when encountering a player and the dialog of a NPC that is supposed to talk, is documented as a script that must be aligned with the rules of the gameplay. *“The NPC script can be just an appendix but you’ve got to read it, so that you make sure your NPC does go off script at the right moment” (P8, ArtGame).*

Third, in the view of those practitioners working in a more technical roles, a MMORPG is – from a SE perspective, a system of systems whose development faces a number of operational complexities and, in turn, entails a myriad of highly technical tasks such as – for example, code optimization, implementing data protocols, service capacity planning, demand forecasting, software performance analysis, and system tuning, all instrumental to getting *“things appear on the screen the way the gameplay demands” (TechGame)*. As per these practitioners, the technical design document included those pieces of technical knowledge that are deemed critical and therefore should be shared among developers. Many of the detailed design choices in this document (e.g. number of simultaneous explosions on screen at ones) are justified based on specific gameplay requirements.

4.1.3. Gameplay as a relationship between players and designers

Eight practitioners (P2, P4, P5, P7, P11, P12, P14, P15) maintained the view of *gameplay as a relationship between players and designers*, which means endless interaction between game authors and players, in which designers provide the choices, the rules, the penalty and reward scoring schemas, and the players use those as recourses available at their disposal to *“generate their individual gameplay experience”* (as one participant put it). Players, hence, engage in *“writing their individual game story lines”* by using their individual approaches to exploring the choices available in the in-game situations they enter. To benefit from those choices, the player is supposed to think of a suitable strategy. For example, ArtGame considers the resources (the series of choices) provided to players as the fundamental building block for forming quests that in turn are one of the ways for structuring gameplay. Gameplay requirements, hence, belong to a quest, that is to a series of choices.

Does the story of the game dictate the gameplay requirements or the gameplay requirements change the story? The eight practitioners found it hard to answer this question. However, based on their first thoughts in attempting to respond, they seem to be divided regarding this question. Three out of the eight perceived the story in their MMORPG projects to pre-determine the gameplay requirements, e.g.

“You [the player] experience the story through the gameplay. So, your gameplay conveys the story.” (P5, TechGame).

The other five practitioners insisted the gameplay requirements generate the story, e.g.

“Content is king! Keeping players playing means rolling out a massive amount of content every once in a while. You give them [the players] the setting and the content, and they will create the story”. (P12, YouGame).

In all three MMORPGs included in our study, the respective MMORPG environment the player with different ways of measuring the outcome or progress of his/her own avatar, but does not hint to any kind of action that the player must take. It is up to the player to decide on the action to take or on the goal to pursue.

While some players would like to follow the goals relatively easily identifiable in the MMORPG world’s structure, others might want to construct their own. Gameplay, therefore ‘emerges’ out of the abilities of the avatar that are provided by the MMORPG-designers and the player’s own abilities to play the game. From RE perspective, therefore, thinking of gameplay as a relationship between game designers and players, implies that gameplay requirements vary along a spectrum ranging from requirements associated to the functions of the player’s avatar to requirements solely depending on the player’s ability (e.g. gameplay relies on an instant strategy, player’s speed and reflexes).

Although there was no agreement on what is first – gameplay requirements or storyline, there was a consensus among our participants that there should be consistency maintained at all times between the storyline and the gameplay requirements:

“It’s the worst feeling that you repeatedly end up where the plot says you should be, no matter how little sense it makes that you should be there.” (P2, TechGame).

“Part of my job is to watch carefully for encounters with enemies in which the power of the enemy and the power level the player are misaligned. A misalignment leads to odd discrepancies where a NPC is way weaker or way more powerful than circumstances or common sense would suggest. Imagine you encounter a giant monster right at the start of the game, yet killing them with ease while you are still at the very first levels and apparently do not have the skills to do so.” (P15, YouGame).

4.1.4. Commonalities, complementarity and balance among the perspectives

Regardless how practitioners conceptualized their understandings of gameplay, they all shared three commonalities:

First, all practitioners were reasoning about gameplay from players’ viewpoint.

“In [MMORPG] design, there is only one perspective, and it’s the player.” (P14, YouGame).

From this viewpoint, gameplay within a level meant providing the player with a set of relevant choices and their consequences. A choice and its consequences are deemed relevant if they are perceived as enjoyable for the player and contribute to the goal of making the player *“want more of the game”*. In the views of the practitioners, gameplay is created not only for but also with the players. In this sense, *gameplay is co-created* where the players are not passive consumers of game contents but also owners of the gameplay as they may decide to use a playing option in a way unanticipated by the game creators. In all the three cases of games in this study, players create and share add-ons and this provides new ways for other players to carry out certain tasks, e.g. casting spells. For example, ArtGame is aware of more than 4000 add-ons available, across 28 categories. These add-ons could be for the purpose of simplification and consolidation of information, e.g. making relevant information more visible while decreasing the amount of less important information. Other add-ons developed by players are in the form of alternative action buttons or could provide alerts for upcoming events in the play process within a level. Also, it’s common for players using end game contents to set up their own statistics-processing tools by writing some code with interface to the online game environment, in order to collect and analyze quantitative information showing how a players group progresses in a challenge. In the experiences of our participants, players use this ‘in-game analytics’ to make their play more efficient:

“Players are informed and concerned about performance and they want to know how far they advanced in killing a monster. 30%? 90%? Imagine you are in a dark cave, too dark that you even do not know where the monster is. You optimize your performance by tracking yours and the monster’s statistics”. (P16, YouGame).

In response to players' activities focused on optimizing their gameplay, one case study company (ArtGames) designed special *training dummies* in all cities in their virtual world, so that players could experiment with their own 'optimized gameplay'.

Gameplay co-creation also extends to the real-life world where players deploy off-game analytics in communities of players' web sites and blogs that serve for brainstorming, sharing experiences and crafting strategies for increasing the players' performance to defeat a NPC (i.e. a dangerous creature). For example, all interviewees agreed that their players participate in opinion polls set up outside the game. As per our practitioners, in co-creating gameplay, players also act as source of inspirational ideas for the designers to consider as candidates for inclusion in the next MMORPG patch/extension:

"Your players can come up with unimaginable approaches to in-game situations. It's shocking to see players' creativity at play... You need to ask our game masters, the helpdesk guys, who are constantly challenged by players who call to say our help desk line they entered situations in ways unimaginable to the game designers..." (P3, TechGame).

"Learning from those players [who play the game in unanticipated ways] is a kind of crowdsourcing for innovation. For example, who could ever have said that players would watch so keenly the avatar's stats and even switch off the features of our beautiful landscape and the sound of the special effects in order to play it more efficiently? I would not tell you the astronomic amount of money this [the sound effects and the landscape effects] all costs!!" (P10, ArtGame).

"Players these days are extremely educated; they do not want just new contents but also some tools to play the game as efficiently as possible. You've got to leverage their ideas somehow..." (P12, YouGame).

An interesting dimension of gameplay co-creation with players, is the players' ability to affect the development of game features that happens after the release of the game. TechGame provided an example of a situation in which players even got actively included into the development process: After release, TechGame noticed that their players did not use the game as the company originally thought. Instead of playing in small groups, the players organized large raids in the game to fight a tough resident evil, which strained the MMORPG environment in an unexpected way, e.g. instead of using their own group channels, players opted for public off game channels to communicate with each other. TechGame collected the gameplay requirements for their next update used by carefully studying what the players did, how they did it and at what point in the monster-fighting process the players used the self-created functionality.

Second, all practitioners agreed on the demarcation of gameplay at lower levels compared to higher levels, and specifically to the endgame levels. In the practitioners' experiences, gameplay applied either to a level in the game, or to the '*end level game contents*', but not to the game as a whole. This understanding reflects practitioners' reasoning about the concept of goal. As client retention was deemed a strategic business goal of the three case study companies, the case study participants agreed that their three MMORPGs were all designed to provide endless playing. The goal of each MMORPG as a system and as a product was *"to keep gearing the avatar so that the player can make"*. Such a goal does not have an end point, nor there is any identifiable place or situation in the game world where it can be reached. From practitioners' standpoint, it is a rather *"ongoing process"* and *"a moving target"*. Because the MMORPG end goal is perceived as too abstract, practitioners consider it pragmatic to reason about gameplay from the viewpoint of a single game level that has its own (pragmatic) goal – namely, to pass the level, and not from the viewpoint of the whole game. In the words of one interviewee:

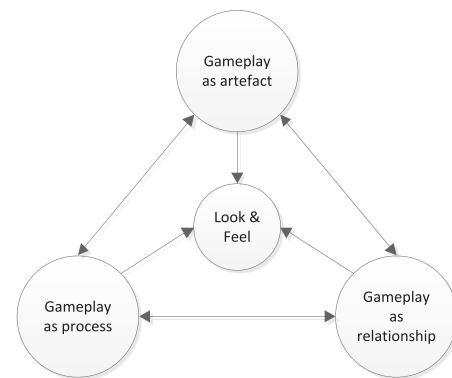


Fig. 5. Complementarity and balance among the three perspectives.

"In a business systems project, you break it down in subprojects and you may have very good reasons to determine specific requirements at subproject level. That's how we deal with gameplay. The only way for you to make meaningful specifications of gameplay requirements is within a level." (P8, ArtGame).

"My job as the level designer is to convey to the developers all the requirements that go to this level. How many quests, what wins would be there for the various classes, what monsters, when you pass the level...these all are my level requirements" (P1, TechGame).

While at lower levels, players may choose individual versus collaborative play mode, for them to be able to cope with the challenges at the endgame levels, they must join a team of players and only collectively they tackle the challenges. In fact, many players go through the levels to arrive as fast as possible to the end level which *"offers a completely different gameplay"* in the sense that its focus is completely on collaboration. Ten out of 16 practitioners referred to this as to the so-called *"group-or-die"* gameplay because the only way for players to continue their play was by joining a group (or raiding guild, in the terminology of the game community).

Third, no perspective was deemed more important than the others. In fact, the interviewers suggested that for a MMORPG to be a well thought our concept, the gameplay requirements need to be treated by all the three perspectives. In the experiences of our interviewees, the perspectives reinforce each other and work together towards creating a common *"look and feel"* (Fig. 5).

4.2. Requirements artefacts dealing with gameplay

All practitioners thought gameplay requirements form a component (*"a chapter"*, *"an essential part with its own heading"*) of the game design document, the pillar of every game development project. In all three cases, the practitioners considered the game design document (GDD) as a living document, in the sense that it's gradually refined following agile project management philosophy. When practitioners refer to it as 'complete', they do not mean 'carved in stone' or frozen, but 'complete' in the sense that it's enough for developing a piece of playable functionality. In the same vein, the gameplay requirements are living requirements and the gameplay requirements document component of the GDD is also a living artefact (see Fig. 6). Its first version is grounded on the game's Look and Feel (LaF) document which in MMORPG projects has the meaning that the project mission and vision documents have in conventional system development projects (e.g. enterprise systems). Each gameplay requirement change is checked for alignment against the statements in the LaF document, meaning that gameplay should support – at all times, the look and feel of the game as envisioned by the game designer. As gameplay requirements are elaborated gradually, traceability links are maintained

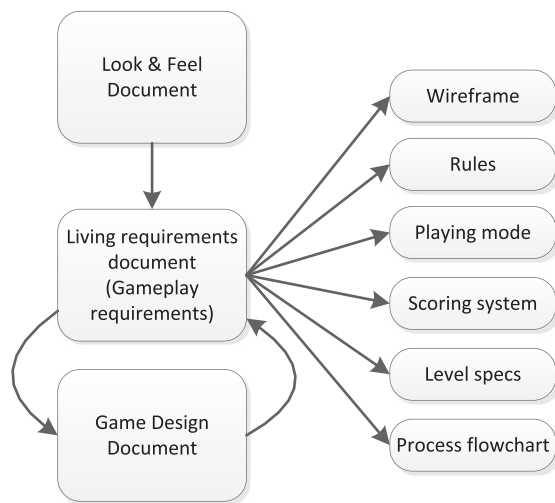


Fig. 6. Composition of the gameplay requirements document as a living artefact.

between the gameplay changes and the affected other parts of the game design. The gameplay requirements document consists of six parts: (1) wireframe, (2) rules of the game, (3) playing modes, (4) scoring system, (5) level specifications, and (6) a process flowchart. The wireframe is the visual blueprint that arranges the game contents for the purpose of achieving the look and feel specified in the LaF document. It can be thought of as “the map of the interfaces”. The concept of rules is self-explanatory (the meaning has already been described in the previous section). The scoring system is the total of awards and penalties. Furthermore, levels specifications explicitly state the design requirements specific to each level in the game. Last, the exact process that the player has to go through within a level is specified by means of a flowchart. The format of the flowchart can vary, for example can be manually made cartoon-like pictures, simple flowcharts based on the built-in Visio stencils, or sophisticated process diagrams using process modelling tools (surprisingly, one interviewer used a business process modelling tool that his company also deployed for documenting business process workflows in accounting and human resource management). Each level’s flowchart is detailed into features. For example, each MMORPG in our case study had a proprietary editor (e.g. a map editor, a layout editor) and its features were listed in detail.

Another distinctive artefact that we observed to depend on gameplay requirements was the so-called ‘paper-prototype’. It is specific to prototyping gameplay requirements at lower game levels (when no code is yet developed). It can take the form of sketches, mock-ups, pencil drawings or more complex artistic drawings. It is therefore a low-cost and easy-to-make way to demonstrate gameplay to the producing company’s employees as well as relevant stakeholders. Gameplay is then experienced and feedback by those who experienced it is collected to confirm gameplay requirements. In all the companies, the colleagues of the professionals who made the paper prototype were the first to “play with it” in order to experience “how it feels to be in the game”. Of course, the very first players to whom a prototype is shown are the team members in the game development project: “It could be everyone that is at work that day; you can ask your artistic director to be a player, or your special effects artist, or a junior or a senior programmer.” (P11, ArtGame).

At higher game levels that feature increasingly more community-based play, the paper-prototype is usually followed by developing a software prototype, an artefact used to demonstrate how playable the gameplay requirements are from players’

community perspective. The practitioners deemed this artefact central to gameplay requirements validation as it meant collecting and analyzing players’ feedback (at community level).

4.3. How do our study participants manage gameplay RE?

All practitioners perceived the gameplay RE process as creative, iterative and non-linear one, in which professionals must be open-minded and be prepared to alternate “fast track lanes with scenic routes” (TechGame). An interesting observation is that no practitioner was able to describe it as a predefined flow of steps that he/she attempted to use in his/her projects. Some interviewees shared that to an external observer this process usually appears as “chaotic”, “judgment-based”, “gut-feel-grounded”. They all meant it as something that does not lend itself easily to a “process description as you do in business process reengineering projects” (P14, YouGame). Instead, they thought of it rather as a ‘system of spaces’ in which spaces delimit various kinds of related activities that together form the continuum of gameplay RE. They referred to them as ‘spaces’ and not process phases or activities, because ‘the spaces’ did not make up any linear structure nor were executed sequentially. While carrying out the activities might look chaotic, in the views of the participants the process consistently brought the targeted results in their projects:

“I’ve never seen a linear, milestone-based process here. We do have a process architecture for gameplay requirements but it’s totally different, just like out of this world... It’s like you have a bunch of tools in your toolbox but without knowing your game level, and its feel and look, you have no way to guess what tool to pool out first; even if you do know the look and feel, and you know the right tool for it, you may find out that it does not fit right in, so you’ve got to tweak it a bit and if it would not work, then go pick another tool. You should try out different things and see what works and then make sure I do more of what works and less of what does not...” (P15, YouGame).

A common concern across all the spaces was the balancing of the gameplay requirements. To practitioners, this meant getting a requirements definition that leads to creating a playable game version that:

- (1) is fair with respect to all players involved,
- (2) brings more pleasure to play as player’s learning and experience advances, and
- (3) is free of repetitive in-game options, which means that no playing option is worthless and the net value of each option is commensurate with the cost of using it.

Our findings suggest that balancing the gameplay requirements is recursive in nature because achieving balanced gameplay with respect to one of the three criteria above may cause an imbalance to emerge with respect to another criterion and this imbalance may need to be smoothened first, before the first one would. As one interviewee said:

“You want to get the gameplay crispy like chips. But chips are fragile, and you’d better treat them that way. The crispier my gameplay gets in one dimension, the more fragile it might get on another, and if this happens then we’d go first to fix the second issue and once is done, then get back to the first. You get back and forth many times until you make it fun to play; fun AND fair, I mean.” (P9, ArtGame).

All practitioners perceived balancing gameplay requirements to be a major challenge in RE for any game, but they deemed it particularly delicate and difficult in case of MMORPGs. In their experiences, this is because players have different entry points in the game and also because the options presented to players, plus the combinations of those options are so many.

“You have to make it fair, regardless how a player choses to set up [his/her] avatar. Part of my job is to make sure we keep stronger players from collecting too many magic trophies and strengths too

quickly. Kind of send them some natural disasters or unfortunate circumstances along the way. And to those poor guys, the newbies, we'd give them hints to group together with someone [so that] they can slay a monster." (P12, YouGame).

To the practitioners, the concept of 'fair with respect to all players' also included achieving gameplay requirement definitions that would not lead to loopholes in the game. A loophole is a combination of choices whose compound effect in the game environment is an important advantage to the player that discovers it. To a player, executing such a combination is a valid series of moves within the game world, however from the game company's perspective it is unintended and highly undesirable because it creates imbalance in the gameplay. Getting the gameplay as 'loophole-free' as possible is therefore a balancing priority for every MMORPG producer. For example, a player should in no way be able to "make easy money" (in the words of two practitioners from ArtGame), which means no one should be able to buy goods cheaply in one server and sell them for a higher price in another. All 16 practitioners in our study experienced loopholes in their projects. From their perspectives, the loopholes could be grouped in two main categories: (1) the so-called "exploits" which may range from errors in the code to software architecture flaws, and (2) "overlooked need for protection against cheating". We found that all three companies included of our study, were assuming the presence of players who would use unauthorized programs on their computers to circumvent the rules in the respective game and gain some advantages. Based on this assumption, they were taking care of put protections in place to prevent unauthorized software from being used while playing and even banning players who don't play by the rules.

Another dimension of balancing that was deemed important by our participants, was related to the necessity to consolidate the needs and wishes of players with huge differences in terms of skills and levels of commitment, e.g. the players at the last level of the MMOG and those that play more casually. As these player meet in the game's world and cannot ignore each other, the concept of *fairness towards each players' group* is perceived of paramount importance. Three practitioners put forward that the key activity in achieving balance is to experiment with alternative sets of rules to figure out which one will reward the lower level players without penalizing the "hard-core" players at the last level. For example, a participant from TechGame shared that in his game project, in order to achieve fairness, they introduced the gameplay requirement of "parking an avatar in a resting place". This means that if a casual player wants to leave the game (because his/her play time is limited and must go somewhere), he/she could bring his/her avatar to a resting place (a garden, a tavern, or a temple) in the game world and still collect some reward points there, while the hard-core players would do "more work and earn respectively many more points".

To cope with gameplay requirements in a way that gameplay balance is achieved, practitioners experienced that they were going through three spaces (Fig. 7). We labelled the spaces 'prototyping', for the activities that lead to creation of paper prototypes or software prototypes, 'play-testing' for activities centered around the discovery of emergent requirements, and 'tweaking' for the gameplay redesigning activities that lead to improving the players' path through the game levels. A tweaking action could be as simple as "tuning system variables' numbers up and down" (TechGame). Or as complex as abandoning a gameplay requirement and replacing it with some new requirements. We note that prototyping and play-testing have the meaning of player-centric exploration and are expected to help discover new requirements, as in MMORPG projects most requirements emerge throughout the game design itself. In contrast, tweaking has the meaning of an intense synthesis of all learning that happened in prototyping and play-testing and use of this synthesis to redesign the gameplay.

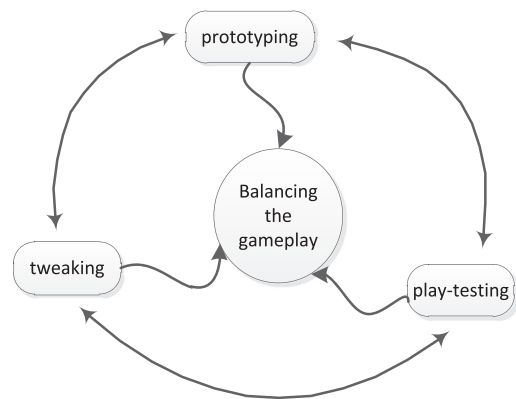


Fig. 7. RE spaces used to balance gameplay.

E.g., in her most recent project the playability analyst at TechGame used prototyping and play-testing as spaces for exploring various amounts of story-telling and full motion video surrounding an end level NPC and a players' guild. Used together, these two spaces led to the creation of up to five new alternative game designs for segments of the game environment (tweaking). Each tweaked design treated differently the players' proactiveness and needs of control. The designs also varied in terms of availability, ease to use and players' cost of the game features included, and these three had to be balanced against their value for the players. The designs were subjected to prototyping and play-testing again whereby prototyping and play-testing were in different proportions regarding each alternative design. These alternating ways of using spaces were continued until the game would be deemed 'balanced' by not only its designers but also by its players (in the targeted community).

We observed however that it was hard for practitioners to pinpoint to one particular space that is more important than the others to the balancing of the gameplay requirements. Instead, they considered all three spaces to be integral to it, and deemed iterations to be the glue that sticks them together. Examples of how striving for balance unites the prototyping, play-testing and tweaking activities was provided by six interviewees. Their reasoning was about using the spaces to gradually and experimentally arrive at the final set of rules that determine the gameplay: e.g. three interviewees worked on the operational rules in projects aimed at delivering end-game contents. Every time, these rules were prototyped and play-tested, they ended up being tweaked as players entered situations unpredictable by the designers and needed new operational rules to get out. The tweaking in itself led to revising some guild rules and adding new add-on rules. Examples of new add-on rules concerned the type of information that a player needed to obtain, which was not obvious from the game world. Plus the format in which it is preferred to be obtained and the timing of its appearance.

How many loops did the participants need through the spaces during a project? Perhaps, it's not unsurprising that no practitioner could give a definitive answer to this question. We found, it was a common experience among the participants that no project team could give in advance any prediction regarding the range in which the number of loops would vary. Some participants worked in teams that made 150 to 200 loops, while others did around 500 loops. Because gameplay requirements are 'living things', the work on them could not be completely finished, but only abandoned. The focus of the RE loops through the spaces was to ensure "a playable enough game is delivered before you exhaust your budget":

"After you make a few loops, you would know your game much better, but even then, you still do not know for sure what you would

get at the end... You need to take care of myriads of visibly small things, that profoundly determine the players' experience, and you just think you brought it close to the finish line, when you figure out, you are a kind of in square one again". (P6, ArtGame).

5. Discussion

This section puts our finding in the perspectives addressed in prior publications in RE for game systems as well as in other fields (e.g. human-computer interaction, game design). Our discussion follows the topics in the previous section. The present discussion builds upon the preliminary discussion points that were described in Daneva (2014). It extends these points by adding the discussion ideas that crystalized at the 2014 RE conference (<http://webhotel.bth.se/re14/>) when the preliminary results of this study were first presented. The present discussion also includes new points that came out of a reflection that happened as part of preparing his paper.

5.1. Gameplay and gameplay requirements

First, our study suggested gameplay requirements are technology independent. An interesting observation is that even those participants who were responsible for implementation (e.g. programmers on server and client sides, web front end engineer) thought of gameplay requirements in non-technical terms, too. This contrasts previously published experiences that found technical specialists take a technical solution perspective and talk a technical language (e.g. Lauesen, 2002). A possible explanation for our observation might be the fact that the implementation professionals in the MMORPG companies are also avid game players (being passionate about the game is a prerequisite for applying for a job in a MMORPG company) and have intimate knowledge of the game as players themselves. So, switching between player's (i.e. user's) and developer's perspective is not that hard as it could be in a case of a 'traditional' system, e.g. an accounting application where the developer is not expected to be a practicing accountant and know the intricacies of accounting rules, and the possible workarounds in the system, from accountant's perspective.

Second, we found three ways to think of gameplay. However, we found that, according to our participants, no particular way of thinking of gameplay requirements was more important than the others. In fact, they all agreed that a good MMORPG should use all the three perspectives. While using multiple perspectives in conceptualizing such a complex system seems sensible and has been a well-established RE practice (e.g. in Kotonya's and Sommerville's textbook of Kotonya and Sommerville, 1989), we think might also give us a hint about the lack of standard definitions of terms in the gaming domain and, in turn, the lack of common understanding in the various communities regarding what gameplay means. The assumption about this lack of standard definitions in RE for games, makes sense as RE textbooks (e.g. Lauesen, 2002) mention, too, that such a challenge is typical for new domains. This, in turn, calls for multidisciplinary approaches to solution development.

Third, our results suggest that gameplay requirements are perceived as functional requirements and not as non-functional ones (quality attributes). We conclude this, because in the experiences of our participants, these requirements were mapped out by using flowcharts or lists of features (e.g., such as those suggested in RE textbooks of Lauesen, 2002 and Kotonya and Sommerville, 1989) which are common techniques for specifying functional requirements. As Section 4.2. indicated, the gameplay requirements document includes many procedural elements (e.g. rules, series of player's actions and actions of NPCs that are documented in flowcharts, among others). In addition to this, we found some

non-functional aspects pertaining to gameplay to be part of the LaF document where **playability** and **fun** are explicitly mentioned as quality attributes. One could assume that these non-functional requirements (playability and fun) are operationalized by means of functional requirements, which are e.g. the rules and the player's and NPC's action flows. In other words, the gameplay requirements can be treated as operationalizations of the quality attributes stated in the Look & Feel document. This position is in line with the reasoning of RE experts on non-functional requirements (e.g. Chung et al., 2004 and Kassab, 2009), which states that in a project, for the non-functional requirements to be technically implementable and verifiable, they need to be decomposed and then operationalized into functional requirements. We did not explore the relationship between the LaF document and the gameplay requirements in this research. Nor we explored how the relationship between playability and gameplay. We though think that such an exploration is a necessary step in order to gain complete knowledge of how the three concepts (fun, gameplay and playability) relate to each other. We, therefore, suggest it as a line for future research.

Forth, we observed that the practitioners were divided regarding the question of whether the story of the game defines gameplay requirements or the gameplay requirements – the story. We think this might be traceable to the two different ways of designing games, top-down (Fullerton, 2008) versus bottom-up (LeBlanc, 2006). This is however a working hypothesis only and needs to be confirmed/disconfirmed in follow-up empirical studies. Moreover, we also assume that, beyond these two game design paradigms (Fullerton, 2008; LeBlanc, 2006), there might well be other contextual factors that might shape the relationship between story and gameplay requirements. More research efforts are necessary to understand the relationship between storyline and gameplay requirements. We so far got hints about the importance and the relevance of such research, as all participants perceived consistency between the two, as an important aspect of RE for game. An important research question for the future therefore could be to understand the types of inconsistencies happening between storyline and gameplay requirements and the practices that game development teams use at the stage of RE to assure consistency at all time.

Fifth, our observations suggest the concept of goal plays a prominent role in reasoning about gameplay requirements. We found, gameplay is bound to a level and as such - to a clearly specified goal. This may mean that gameplay requirements would naturally lend themselves to the goal-oriented RE (GORE) approaches. Our practitioners however have shown no awareness of the deployment of these approaches in the MMORPG industry. As part of this study, we made a search in the Scopus database (www.scopus.com) for empirical studies that used GORE in game context. We found the 2017 work of Miguel Angel Teruel (Teruel, 2017) as the only research publication on evaluating the applicability of GORE to collaborative games (including MMORPG). This author proposed a modified GORE approach to collaborative games and their results demonstrated its suitability to the game domain and the potential benefits of using GORE. Although these authors' case study is on a game that is much smaller in scale than the games in our case study companies, and their results might not be generalizable to our practitioners' settings, we think that the ideas put forward by these authors are indicative and warrant further research and evaluation in light of MMORPG context. Further empirical work to evaluate the fit of GORE and MMORPG projects is necessary and worthwhile because it potentially could reveal how existing RE methods could be leveraged to this new domain. Such studies would provide the solid evidence needed by both practitioners and researchers on the benefits and limitations of GORE in this particular context.

Sixth, we found gameplay requirements are co-created with players and that game development projects use player-generated and tested ideas to extend their MMORPGs. This phenomenon could possibly be explained through two different but related concepts: first through the concept of crowdsourcing-based RE, which is about RE practices leveraging freely available information resources of the crowd (the players, in this case). Second, through the concept of ‘user innovation communities’ (Verganti and Pisano, 2008), that are networks “where any user can propose problems, offer solutions and decide which solution to use”. Our study suggests that the players of MMORPGs are actively forming such communities and that the three companies are well aware of the business benefits this brings. Numerous blogs, wikis, and level-specific forums are created and maintained voluntarily by players of the games produced by our three case study organizations. That these communities have influence on the gameplay requirements was clear to our case study participants. For example, staff of our companies regularly scan players’ online sites to understand the bugs and loopholes players discover. This understanding is the input to the gameplay requirements process for the next release of the game. However, what is the full range of implications these communities have for MMORPG RE is hardly known and seems an interesting line for future research. We consider this question important because recent empirical studies (Crenshaw and Nardi, 2016; Crenshaw et al., 2017) in MMORPG suggest that the collaboration between game companies and players is not always easy. In fact, a 2017 study of Crenshaw and colleagues (Crenshaw et al., 2017) on the social experience in World of Warcraft signals a growing trend in the MMORPG area which suggests that players valuing the social experience in earlier game versions, severely disagree with the game developing company (Blizzard in case of World of Warcraft) and create alternative PvP game servers, also known as “private servers” where they can experience the old version of the game and the social experience that goes with it. These private servers allow players to return to previous versions of a game before changes that modified it (Crenshaw et al., 2017). All three case study organizations indicated their awareness of the existence of private servers concerning their games, however the importance of these users and alternative servers for future game RE was unknown.

5.2. Gameplay artefacts

Our study revealed some linkages between the gameplay requirements and a number of other documents created in MMORPG projects. First, we found that gameplay requirements take a prominent position in the game design document. This agrees with the studies of Callele (2005) and Alves (2007) regarding the fact that gameplay and game design are closely linked concepts. However, while Callele (2005) presents gameplay as an artefact on its own, preceding the elaboration of the game design, our study found that (1) the LaF document precedes the process of gameplay requirements development and (2) that the gameplay requirements do not precede the game design, but are an integral part of the game design. In other words, gameplay requirements and game design are co-evolving together. Our observations that gameplay requirements form living documents and are created by using agile development philosophy agree with textbooks in game design (e.g. Fullerton, 2008).

Second, the gameplay requirements are linked to the NPC script (artistic document), and the technical design document (technical SE document). While the presence of relationships between artistic and very technical documents is not surprising in itself, open question appears to be regarding the traceability management among these documents. The mutual adaptation of the gameplay requirements to the NPC script and to the technical design document, and

vice versa, assumes the presence of coordination, communication and consolidation which so far has been under-researched in the RE literature. Further research on these processes and the mechanisms that make them efficient warrants further attention.

Third, we considered the joint co-creation of gameplay requirements by game designers and players (organized in communities, or guilds) as one of the most insightful findings in our study. While the designers create the choices and the consequences for players taking these choices, the players are inventive in the sense that they (1) create rules of play that are not intended by the designers, (2) instill codes of conduct, (3) impact how the game is facilitated, and (4) impact how development plans would go next. This finding agrees with the study in Nardi (2010) that found gameplay as socially negotiable, which means that players and game developers together determine the rules. One could also think of parallels in meanings between our concept of gameplay co-creation and the concept of player-managed gameplay identified in the survey-based study of Deaker (2012) of the World of Warcraft, a leader in the MMORPG market.

Our finding that gameplay is co-created with players using in-game and off-game analytics, could be explained with the proactiveness of the players and their willingness to form groups. As Ducheneaut et al point out (Ducheneaut, 2006), the end level gameplay experience (of World of Warcraft, in case of Ducheneaut, 2006) is produced through guild-centric collaboration, exclusively. Furthermore, our results make us think that the intensity of co-creating gameplay seems to grow with the advancement of the levels at which players play. According to our participants most blogs and in-game and out-of-the game analytics tools were known to be created by players identifying themselves with the end level of the respective MMORPGs included in our study. We however consider this a working hypotheses only and future research is needed to collect possible evidence in its support.

5.3. Our practitioners’ managing gameplay RE

Our findings suggest that there is no common process that companies call “the right way” to manage RE for MMORPGs. We conclude this from the observation that the case study companies had ‘spaces’ that collectively helped them deliver the requirements. They did not point out to any process or pre-defined sets of steps. We note that we preferred the concept ‘spaces’ over ‘steps’ because in the experiences of the practitioners they usually were not done sequentially. E.g., gameplay requirements may loop back through prototyping, tweaking, and play-testing more than once as the game development team refines them and explores new directions. This agrees with Kasurinen (2014) regarding the role of prototyping. However, our finding seems to disagree with Callele (2005) that prototyping is hard. In fact, we found that the paper-prototype was a high value/low cost way to cope with gameplay requirements.

Our understanding of managing gameplay RE by using spaces is in line with the creative processes studied in the field of design, e.g. Brown et al. (2008). We think this is unsurprising as gameplay requirements definitions result out of the creativity of multidisciplinary teams bringing a broad range of backgrounds. We however notice that no RE research has applied so far the theoretical lens of design thinking to the study of RE phenomena in the gaming industry. For this reason, we think it would be interesting to investigate the commonalities between managing gameplay requirements and design thinking approaches (Brown et al., 2008). This forms a line for future research.

Thinking of the RE spaces presented in Fig 4, one could argue that each space may well include game development activities, which collectively “engineer” fun through continuous experimental

tion and therefore it might not be useful to consider separating RE from these development activities. This reasoning would be consistent with the agile paradigms of software development and their applications in game projects (e.g., as presented by the long time game development expert Clinton Keith in Keith (2010)). However, to understand the nature of gameplay RE and its relationship to game design, more research is needed.

Balancing the gameplay requirements seems to be a consolidation process in which individual player's perspective is gradually synchronized with the players' community perspective. While this might sound logical – knowing the sociability of MMORPGs, our findings suggest that it is far from straightforward. For example our findings suggest that one of the most difficult things in the process of achieving loophole-free gameplay requirements, is collecting requirements from individual player's perspective (e.g. rules within a level, rewarding points and penalty points, applicable to each player) and then consider the benefit for the entity also in a social context.

6. Limitations

We used the checklist in Runeson and Höst (2009) to assess the possible threats to validity in this study. As it is exploratory, the key question to address when evaluating the validity of our results, is Yin (2008): to what extent can the practitioners' experiences in gameplay RE be considered representative for other MMORPG companies and other (non-MMORPG) game producers? While the case study companies are not representative settings for all the possible ways in which gameplay RE can happen, following Seddon and Scheepers (2011) we could think that it might be possible to observe similar experiences in MMORPG projects and companies which have contexts similar to those in our study, e.g., other MMORPG producers who use the subscriptions-based business model and have incentives for designing gameplay that serves the purpose of player retention, and has no end goal to achieve. This subscription model requires companies to commit to high quality standards (so that they retain their player base as long as possible) and incentivizes them to come up with steady releases of content. Such MMORPGs organizations might experience observations similar to ours.

Second, the three MMORPGs in this study are real-time strategy games. It might be well possible that our findings hold for this specific class of MMORPG and the findings might vary if we would replicate this study in the context of massively multiplayer online sports games (such as e.g. Baseball Mogul Online), massively multiplayer online racing games (such as e.g. Kart Rider, Upshift StrikerRacer), massively multiplayer online dance games (e.g. Just Dance 2014), or massively multiplayer puzzle games that are based entirely on puzzle elements. Further research with participants from companies producing such MMORPGs is necessary in order to understand better the extent to which gameplay requirements are conceptualized in similar and different ways compared to real-time strategy games.

Third, would our findings hold for other types of game systems, e.g., video games that are free or that are for a single player? One could possibly assume that requirements and design choices might be highly influenced by the business model (in this case, subscription-based versus based on the concept of “free” (Anderson, 2009)) adopted by a game producing company. This assumption rests on a finding from a 2014 study (Daneva et al., 2014) co-authored by the author of this paper, which suggested that the way in which practitioners engineer quality requirements is traceable to incentives created by the underlying business model of organizations. For example, the finding that gameplay requirements result out of the interaction between players and developers, may

well be explained with the subscription-based model and the social community-building component that is part of it. Of course, to know for sure if this is the case, more empirical research is needed to be carried out in the future.

As part of reflecting on the question about how our findings might be different in the context of video games for a single player, we looked at video games in general and at games that run over social networks such as Facebook, in particular. The category of video games refers to any game with a screen/display component made possible by the use of a computer, or computer-like device (a console such as the Playstation, Xbox 360), or a mobile phone. This is a general category which includes games designed for one player only – the so called ‘player versus computer’ type of games (e.g. Angry Birds or Terraria) but the category is not limited to such games, and it includes also MMORPG. Video games that are designed for one player are more often than not highly accessible designs that put less emphasis on competition and skill building. They offer in-game situations that demand modest skills and focus on rewarding and encouraging players to keep them engaged. We think that as the concept of gameplay is essential to any type of gaming systems (Fullerton, 2008), it might well be possible that a part of our findings would be observable in the contexts of any game that is structured in levels, e.g. gameplay as an artifact (a set of rules, penalties and awards) is likely to apply to any game. Therefore, gameplay as a process and as an artifact might well be conceptualized in the same way as it was by our case study participants. However, we think that requirements co-creation of gameplay requirements by designers and by players might not be realistic in case of standalone games (such as Angry Birds, or Terraria). Next social games are a sub-category of video games facilitated by a social media platform (e.g. Facebook) and involving playing with other subscribers to the same social media platform, either in a PvP environment or cooperatively in a PvE sense. As many of these games are free, require no installations and are relatively easy to get started, they do not have any special demand to skill level and learning time commitment. The accessibility of these games makes us assume that aspects of our findings concerning balancing would not be applicable. For example, the challenge to consolidate hard core gaming experience with the experience of casual game players would not exist in such context. Of course, to know for sure, we need to spend more case study research efforts in these particular settings.

Furthermore, as part of reflecting on generalizability of our results, we considered the context of serious games. If we replicate this study in serious games contexts, e.g. those described in the systematic review on serious MMORPGs that teach leadership skills (Lopez et al., 2013), then we expect the results to be different since pedagogical goals and learners' behavior models dominate the game design and impact the choices and the time available to players.

Fourth, although our participants were similarly distributed across the three companies, the study is limited to the participants' professions and their roles and responsibilities in the companies – be it on the software development side (e.g. software engineers and project leads) or on the artistic sides (e.g. artists responsible for the characters, the sound and the visual effects). For example, we have not included business roles such as marketing, producing director, or production managers on the publishing side of an MMORPG business. It might be therefore possible that professionals in these resorts might add new perspectives on gameplay requirements.

Last, we also acknowledge three inherent weaknesses of interview techniques (King and Horrock, 2010 Norris, 1997) that we accounted for in this study. The first is the extent to which the participants answered our question truthfully. We attempted to minimize this threat by involving volunteers, under the assumption

that if a practitioner would not be able to be honest, he/she could decline his/her participation at any stage of the research. Second, in any interview study, there is always a residual risk that the respondents were extremely selective in choosing the documents for sharing with the researcher and that the respondents might have incentives to share the very best of their work. Third, it is possible that the researcher might instill her bias in the data collection and data analysis processes. For example, it might be possible that the affinity of researcher with certain interviewees, games, data, theories, concepts, and explanations, interfered with data analysis. As Norris indicates, no qualitative research is immune against these biases (Norris, 1997). In this respect, we resorted to the practices suggested by Yin (2008): (1) we included participants with diverse jobs (14 different positions and roles) and with diverse contributions to the gameplay RE process; this allowed the same phenomenon to be evaluated from diverse perspectives (data triangulation (King and Horrocks, 2010)); (2) the interview answers were sent to each participant prior to data analysis to confirm the information he/she provided; and (3) we had the data analysis report reviewed by 10 of the 16 interviewees (some of whom provided feedback on clarifying the concepts, but this did not change the analysis). Following Norris, we also used the practice of introspection regarding what framed the researcher's interpretations of the world: the researcher acknowledges her background as a professional consultant in RE for large-scale enterprise resource planning software projects, a career that took place prior to joining the university (in 2004). She also acknowledges her use of coordination theory and organizational theories in her past (2006) research. This professional experience might have influenced the researcher in her thinking of gameplay requirements from a process perspective. However, to reduce this bias, the researcher discussed these points with those interviews (P3, P6, P9, P10, P16) who put forward the process perspective. This served the purpose to re-evaluate the researcher's impressions of respondents and to challenge any pre-existing assumptions on the researcher's side. As part of these conversations, four interviewee (P3, P6, P10, P16) provided examples of gameplay documents and diagrams that illustrated what made them think of gameplay requirements as a process. This was a form of data triangulation, an anti-bias measure recommended by Yin (2008).

7. Implications

Our findings carry important implications for researchers, for practitioner, and for teachers in Requirements Engineering, Software Engineering and other related fields included in the Computer Science education. This paper's discussion on the research implications extends our very first reflection reported in Daneva (2014). In contrast to Daneva (2014) where we jotted down some very specific follow-up RE research actions, in this paper, we contribute a more extensive elaboration of gameplay RE research lines for the future based on a comparison with very recent publications in the field. We add a new reflection on the possible takeaways from our work for practitioners and for teachers.

7.1. Implications for RE research

Our study has two types of implications for RE researchers: from a RE contents perspective and from a research methodology perspective.

7.1.1. Implications concerning contents of RE research

First, our discussion (Section 6) presented a number of possible lines for future research. We observed that our findings in gameplay RE did not call for development of new methods. Instead,

knowing what we know now, it seems to us that more evaluation work would be an important thing to do first. This is line with the recent work of Miguel Angel Teruel (Teruel, 2017) on the evaluation of GORE to the game development context.

Second, we found RE spaces are repetitively and iteratively used for balancing gameplay. Using any space includes some form of formulating assumptions about what the gameplay requirements should, would and could be, and then checking the validity of those assumptions. One might therefore say that the final set of gameplay requirements comes out as the sum of 'mini-validation' work taking place in each of the three spaces. No one among our interviewees talked about gameplay requirements by using the requirements validation terminology that we know from RE textbooks (e.g. Lauesen, 2002 and Kotonya and Sommerville, 1989). Understanding what gameplay requirements validation means and how it happens, therefore seems a promising line for future research.

Third, gameplay requirements are only one piece of the puzzle in RE for games. In the MMORPG area, we found that these requirements are linked to the Like & Feel document, the storyline, the NPC script and some very technical design documents. Traceability management between these artefacts has hardly been addressed, which we think indicates a relevant area for research in the future.

Fourth, we found that gameplay requirements are not equal to playability or to players experience requirements. Our results however give hints that dependencies exist between gameplay, playability and players experience. To understanding the nature of these dependencies it is necessary to execute more case studies with practitioners in the field.

Fifth, we checked our results for possible relationships that might exist between the occupational background (i.e. the jobs) of the practitioners and the ways in which they perceived the gameplay requirements. For example, one might assume that project leads could perceive gameplay requirements as an artefact while practitioners with more 'creative' jobs might perceive them as relationship between players and designers. However, our findings do not indicate the presence of a relationship between occupation and perceptions. This – of course, might be accidental. Or it might indicate the presence of other practitioners' characteristics that could possibly explain why we observe what we observe. In any case more research is needed to elucidate this.

Sixth, in the field of game design there is a growing community that develops game patterns as a foundation for reuse in the process of designing games. Similarly to other software engineering sub-areas where patterns have been used, game design patterns are considered simple collections of reusable solutions to solve recurring problems in game development projects. This community suggested some ontology-based techniques to game design, such as those in Zagal and Bruckman (2008) and Falstein (2004). Because these technique leverage the benefits of common game vocabulary and also because ontological frameworks have traditionally been used as foundation for RE technologies (e.g. in GORE (Guizzardi et al., 2012)), we think that it might be worthwhile adapting the game- pattern-based techniques to serve RE purposes. Provided the existing amount of established game design patterns (around 300 (Bjoerk and Holopainen, 2005)), an interesting line for research could be to collect a catalogue of requirements patterns that meet the special demands for the development of MMORPG systems and are induced from those established patterns.

Last, the three ways to conceptualize gameplay could be subjected to a more comprehensive survey with participants from a large number of companies. If confirmed, our findings would have broader implications to RE (e.g. we may need to design RE tools that support this multiple conceptualization view).

7.1.2. Theoretical and methodological implications

First, this study produced a substantive theory of gameplay requirements and their RE spaces. Our analysis went deeply into 'the salient attributes of phenomena' as Gregor (2006) (Gregor, 2014) calls it. The substantive theory is descriptive in nature. It did not pretend to claim that the three perspectives (process, artefact and relationship) are the only ones that may be possible. Nor we strove for agreement among our participants while analyzing our data (as R.Yin indicates (Yin, 2008)), exploratory qualitative studies help a researcher develop a deep understanding of a phenomenon by considering as many viewpoints as possible). According to Gregor (a research methodologist on theories and theory-building), such an analysis is of particular benefit to other researchers in research contexts where very little knowledge is available about the phenomena of interest (Gregor, 2006, (Gregor, 2014) p. 623). In line with Gregor's reasoning, despite the fact that we do not and cannot provide a definitive description of the ways in which game producing companies and practitioners working in them reason about gameplay requirements, we made an important first step towards making knowledge in this area explicit. This knowledge could be helpful in the future to work out which replication studies are desirable as well as to discover new and relevant research questions. We elaborate on this in Section 6 where we suggested lines for future research.

Second, the layered meanings of the gameplay concept makes us think that it would lend itself to multidisciplinary empirical research in which researchers could possibly use existing theories from other disciplines, e.g. social psychology and media communication, to explain gameplay in more depth. E.g. our study indicated the joint co-creation of gameplay requirements by game designers and players in the MMORPG domain. To understand the internal working behind the relationship between designers and players in RE, we think that future research could leverage theories about client-vendor value co-creation (e.g. Östblad et al., 2014) from the area of e-commerce. MMORPG are in fact examples of e-commerce and therefore would be a suitable area for the application of these theories for the purpose of understanding gameplay requirements. Furthermore, applying the theoretical lens of design thinking (Brown et al., 2008) could aid the exploration of those collaborative RE processes that yield gameplay requirements.

Third, the MMORPG world of players interacting online gets increasingly more distant from the traditional lab settings we know in RE-experiments. As MMORPG environments keep track on a broad range of statistics and each player generates huge volumes of data all being recorded in the game system, an empirical approach to investigating gameplay as it happens real-time, could be the direct observation method. A researcher might consider entering the game, creating his/her own avatar and then collecting observations of how social gameplay works, while playing the game. E.g. a researcher can evaluate theoretical models of coordination and collaboration in the context of social networking in online game communities. Examples of this research approach are provided in Gross (2012), Castronova (2013), and Warmelink and Sitonen (2011). Leveraging these experiences in the RE field is a line for future research.

Fourth, MMORPG maintain free repositories with reviews that could be used by researchers as sources of qualitative information that could be further analyzed. These sources could be considered in a multi-method research design where they could be combined with other data collection techniques in and outside the game. For example, Zhu et al. (2017) recently demonstrated how to leverage these huge repositories of players' comments towards evolutionary requirements for games. Methodologically, such approaches based on big data are considered good candidates for inclusion in a multi-method research process that could be deployed in empir-

ical studies to evaluate the generalizability of the findings in our research.

7.2. Implications for practitioners

This study has actionable implications to RE practice. To RE specialists who look for opportunities to join the game industry, our research indicated that the players' perspective is the one driving gameplay RE. To get intimate knowledge of the players' perspective, it seems unavoidable for RE specialists (who want to switch to the game industry) to become MMORPG players first. In contrast to other domains, e.g. accounting, health-care, where RE professionals do not have to be actual users of the systems in order to be able to understand the world of the users sufficiently well and do decent job, this study revealed that being a player seems a prerequisite for being a good gameplay requirements engineer. In a world where the product (the game) is produced by people with drastically diverse background, as Callele et al put it (Callele, 2005), being a passionate game-player may well be the only commonality that sticks professionals together.

Similarly to other RE practices whose usefulness is contingent on the contextual settings of the project where these practices would be applied to, we could imagine that some perspectives on gameplay requirements might be more relevant than others in a particular context. For example, if a game project deals with an advanced level or with an end level, then the gameplay-as-a-relationship might be more relevant than the other two perspectives just because of the high level of sociability in the MMORPG at that level. In the same vein, if a project produces an extension of a lower level functionality, then it might be better to frame gameplay requirements from process perspective.

To RE method and tool vendors, our findings that gameplay requirements are perceived as functional requirements and that traditional flowchart and process modeling techniques are a good fit with gameplay modelling mean that there could be a new market segment just waiting to be explored. To the best of our knowledge no RE tool vendor visible in the RE community has geared their tool to serve MMORPG companies. This study suggests that MMORPG companies could become potential users of such tools, i.e. they form a market and maybe tool vendors might be interested in exploring and developing it further.

Our findings suggests that play-testing in MMORPG at the level of requirements happens without using sophisticated testing tools, but 'paper prototypes'. This is very different from development of e.g. enterprise information systems where various testing processes are automated and systematically supported by means of commercial testing tools. Being a play-tester for early gameplay requirements therefore requires no technical expertise in terms of tools, or testing technology, but willingness to play the game under development and ability to communicate the own playing experience as suggestions for changes in the requirements. This is a very different situation compared to the project setting assumed by commercial testing tool vendors. As MMORPG developing companies seem to be a very distinctive market, testing tool companies might need to redefine their offerings should they want to service them.

7.3. Implications for RE education

This study has some implications for RE educators. First, our findings suggest that gameplay requirements are different and are conceptualized, elicited, and validated differently compared to 'conventional' projects, such as enterprise systems. Therefore, it might be sensible for RE teachers to consider that game development projects would call for different RE skills compared to the skills taught by using well-known RE textbooks (e.g. Lauesen, 2002). This point concurs with Valente et al. (2017)

and Murphy-Hill et al. (2014). Murphy-Hill et al. (2014) who also thought that teachers should expose students interested in careers in computer games, to different software engineering methods compared to those taught for ‘conventional’ projects. For example, if a paper prototype is a quick and inexpensive way to check how a set of gameplay requirements matches the Look & Feel of the game, then students would need some exposure into the realm of creating paper prototypes, e.g. Snyder (2003), for fast or just enough gameplay requirements validation.

Second, RE teachers could possibly leverage techniques that they might have been using in students’ projects in other domains, such as social robots and assistive adaptive technology, e.g. for patient care. For example, Sullivan’s and Smith’s lessons of teaching game design in Sullivan and Smith (2011) suggest a number of theories from those domains to be included in teaching. In the same vein, we think that RE teachers might consider including some relevant theories for RE, such as LeBlanc’s Mechanistic-Dynamics-Aesthetics framework (LeBlanc, 2006). Borrowing techniques that are applied in social robotics systems and adaptive technology projects could be a meaningful addition to the ‘traditional repertoire’ of RE techniques taught in many universities (based on e.g. Lauesen (2002) and Kotonya and Sommerville (1989)).

Third, because the RE spaces hint to similarities with the design thinking approach used e.g. in service design, it seems to make sense to adopt and adapt some teaching and learning practices (Brown, 2009) from Design Schools, e.g. the Hasso Plattner Institute of Design at Stanford (<http://dschool.stanford.edu/>). This would extend the instrument choices in the toolbox made available to RE students.

Fourth, in teaching RE courses there is a common understanding in many universities that the RE teacher teaches RE techniques assuming students have no business domain knowledge. Case studies are therefore used to provide meaningful contexts in which the students would apply the techniques taught e.g. as in Lauesen (2002) and in Scacchi (2017). In the MMORPG development context, however, ‘domain knowledge’ is acquired by playing the respective game. Assuming a RE teacher wants to embark on a course in RE for game projects (in particular MMORPG projects) and his/her audience has years-long MMORPG playing experience, then the teacher should be prepared to invest a certain amount of hours in “acquiring the domain knowledge” so that he/she and the audience have a shared understanding of the settings where the RE techniques would be exercised. The time that this demands is significant (Ducheneaut, 2006), which might pose a serious challenge to those teachers who have no prior knowledge of MMORPGs and who must balance demanding job priorities in their careers. Teachers therefore might find themselves better-off to collaborate with game development companies in course development and include practitioners as lecturers in their courses in order to provide not only a meaningful contexts for the students to study but also an in-depth knowledge of the game RE process in general.

7. Conclusions

The attention to game RE and gameplay requirements has been growing in recent years. A few conceptual works have posited that understanding gameplay requirements from game designers and from players’ perspective is critical to our understanding of the field. We sought to better understand this commonly observed but understudied class of requirements in the context of one particular class of large-scale game systems, namely MMORPG. Our exploratory study explicated what practitioners in MMORPG development teams think of gameplay requirements and how they manage gameplay RE. We uncovered multileveled meanings of the gameplay concept – as a process, as an artefact, and as a relationship between players and game designers. What is more, we found that

these perspectives are equally important and complimentary. This has important implications for MMORPG RE. First, it seems this makes it hard for companies to come up with a common process for eliciting and engineering gameplay requirements. Instead of setting up RE processes, we found that practitioners dealt with RE spaces (and not steps) aiming at collectively addressing the key concern in MMORPGs: *balancing* the gameplay. This balance-driven requirements process is co-created by game developers and players, is endless within the MMORPG, and is happening both in-game and off-game. It heavily relies on experimentation which entails alternating ‘paper-prototyping’ and play-testing for the purpose of gameplay requirements validation.

We compared and contrasted our findings with results from previously published research (up to March 2017) and drew some implications for researchers and theorists, for practitioners, and for RE educators. We hope this study opens up further discussions and advances theory to generate a more holistic, comprehensive understanding about the nature of gameplay requirements and their role in engineering complex, large-scale game systems. We conclude that in order to further the research and education in this area we need a close collaboration with game companies, not only in the MMORPG sub-area but also in other game sub-areas such as social games and standalone video games. Multiple case studies are needed to compare and contrast processes for engineering gameplay requirements across these diverse contexts. Only then, we could come up with relevant theories for game RE, with educational case studies (such as those put forward in Scacchi, 2017) and with pragmatic and evidence-based advice to practitioners on which RE technique to use in what game context.

Appendix A. Interview questions

1. Introduction

“Please consider a concrete project where you were involved in work on gameplay requirements. It could be your last project, or any recent project at your company. We will first discuss some general characteristics of gameplay requirements, and then maybe we would use some examples from your project.”

The interview includes three sections that will be covered in a semi-structured fashion. The questions should be seen more as means to get a conversation started, and the line of questioning should not be considered to be too rigid.

Section 1. Gameplay requirements

Q1.1: Please tell me about your tasks that deal with gameplay requirements?

Q1.2: Thinking of your experiences at [name of company], what are gameplay requirements for you? How would you define them? What’s unique about them?

Q1.3: When you talk to other fellow project members, what is it in the gameplay that you talk most of the time? Why is this the subject of your conversations so often?

Q1.4: Is there anything in the gameplay requirements that is particularly hard to achieve?

Q1.5: Why was it hard?

Q1.6: What part of this difficulty did you resolve yourself?

Q1.7: Who do you collaborate with, when dealing with issues concerning gameplay requirements?

Q1.8: Could you give an example, maybe from your last project, when you had setbacks and how you sorted this out? (**concrete case**)

Q1.9: Is there any part in your work on gameplay requirements that you consider critical? Why is it critical?

Section 2. Documenting gameplay requirements process

Q2.1: Who do you work with in regard to the gameplay requirements? How do you share the requirements with that person?

Q2.2: Regarding requirements documentation, is there any specific approach that you use in your practice? How do you go about documenting the gameplay requirements?

Q2.3: Is there any specific template, or tool that you use? Any modelling notation?

Q2.4: Are there any dependencies between gameplay requirements document and other documents that you use in your project? What are those?

Q2.5: If a gameplay requirement is poorly defined or is missing, to whom this would matter? Why?

Q2.6: What in your experience, is the most difficult part of documenting these requirements?

Q2.7: Where did updates come from?

Q2.8: Who makes sure the rest of the documents are consistent? How does this happen?

Section 3: The life of gameplay requirements

Q3.1: How the gameplay requirements come to live? Where do they originate? And who is responsible for them?

Q3.2: And how did this happen in your project?

Q3.3: Are any processes used to help get the requirements right?

Q3.4: Could you describe the process that you observe being used in your last project?

Q3.4: Who provides feedback on gameplay requirements and what actions are taken based on the feedback? (**concrete case**)

Q3.5: Who receives the feedback? Who decides on the actions to take? (**concrete case**)

Q3.6: At what point do you think of the players in regard to gameplay requirements? (**concrete case**)

Q3.7: How do you go about validating gameplay requirements? Could you give specific examples from your experience?

Q3.8: How did it happen in your project? (**concrete case**)

Q3.9: How did you know when to stop with validation efforts?

Q3.10: Tell me more about the players' feedback. Do you have cases when the players' feedback overrides the gameplay requirements that you dealt with so far? If so, could you give examples of what exactly happened and why it happened.

References

- Alves, C.F., et al., 2007. Challenges in Requirements Engineering For Mobile Games Development: The Meantime Case Study. *RE*, pp. 275–280.
- Ampatzoglou, A., Stamelos, I., 2010. In: *Software Engineering Research For Computer games: A systematic Review*, 52. *IST*, pp. 888–901.
- Anderson, C., 2009. *Free: The Future of Radical Price*. Hiperton, NY.
- Bainbridge, W.S., 2010. *The Warcraft Civilization: Social Science in a Virtual World*. MIT Press.
- Bell, J., Cooper, K.M., Kaiser, G., Sheth, S., Report from the second international workshop on games and software engineering (GAS 2012), 37 (6), 2012, 1–6.
- Bensabat, I., et al., 1987. The case research strategy in studies of information systems. *MISQ* 11 (3), 369–386.
- Bergstrom, K., et al., (2010). Exploring aesthetical gameplay design patterns, *MindTREK* 17–24.
- Bjork, S., Holopainen, J., 2005. Patterns in Game Design. *CRM*.
- Boyle, E.A., Connolly, T.M., Hainey, T., Boyle, J.M., 2012. Engagement in digital entertainment games: a systematic review. *Comput. Hum. Behav.* 28 (3), 771–780.
- Breckenridge, J.P., Jones, D., Elliott, I., Nicol, M., 2012. Choosing a methodological path: reflections on the constructivist turn. In: *Grounded Theory Review*, 11. Sociology Press, pp. 64–71. June.
- Brockmyer, J.H., et al., 2009. The development of the game engagement questionnaire: a measure of engagement in video game-playing. *J. Exp. Social Psychol.* 45 (4), 624–634.
- Brown, T., 2009. *Change By Design: How Design Thinking Transforms Organizations and Inspires Innovation*. Harper Business.
- Brown T., *Design Thinking*, HB Review, June 2008, 84–95.
- Callele, D., et al., 2005. In: *Requirements Engineering and the Creative Process in the Video Game Industry*, 05. *RE*, pp. 240–252.
- Callele, D., et al., 2009. Augmenting Emotional Requirements With Emotion Markers and Emotion Prototypes. *RE*, pp. 373–374.
- Callele, D., Dueck, P., Wnuk, K., Hynninen, P., 2015. Experience requirements in video games definition and testability. In: *RE*, pp. 324–333.
- Caroux, L., Isbister, K., Le Bigot, L., Vibert, N., 2015. Player–video game interaction: a systematic review of current concepts. In: *Computers in Human Behavior*, 48. Elsevier, pp. 366–381.
- Castronova, E., et al., 2013. Designer, analyst, tinker: how game analytics will contribute to science. In: *Game Analytics, Maximizing the Value of Player Data*. Springer, pp. 665–687.
- Charmaz, K., 2003. Grounded theory: objectivist and constructivist methods. In: Denzin, N.K., Lincoln, Y.S. (Eds.), *Strategies of Qualitative Inquiry*. Sage, pp. 249–291.
- Charmaz, K., 2007. *Constructing Grounded Theory*. Sage.
- Chung, L., Nixon, B.A., Yu, E., Mylopoulos, J., 2004. *Non-Functional Requirements in Software Engineering*. Springer.
- Cooper, K.M.L., et al., 2014. Towards model-driven requirements engineering for serious educational games: Informal, semi-formal, and formal models. *REFSQ* 17–22.
- Cornett, S., 2004. The usability of MMOG: designing for new users. In: *SIGCHI Human Factors in Comp. Syst. ACM*, pp. 703–710.
- Crenshaw, N., LaMorte, J., Nardi, B.A., 2017. Something we loved that was taken away": community and neoliberalism in world of warcraft. *HICSS*.
- Crenshaw, N., Nardi, B.A., 2016. It was more than just the game, it was the community": social affordances in online games. In: *HICSS*, pp. 3781–3790.
- Daneva, M., et al., 2013. Agile requirements prioritization in large-scale outsourced system projects: an empirical study. *JSS* 86 (5), 1333–1353.
- Daneva, M., 2014. How practitioners approach gameplay requirements? An exploration into the context of massive multiplayer online role-playing games. In: *RE*, pp. 3–12.
- Daneva, M., Marczak, S., Herrmann, A., 2014. Engineering of quality requirements as perceived by near-shore development centers' architects in Eastern Europe: the hole in the whole. In: *ESEM 2014*, pp. 190–200.
- De Heij, B., et al., 2013. *The Global Games Market: Newzoo Trent Report*. Newzoo Inc., Green Bay, USA.
- Deaker, C., et al., 2012. A Survey of Players' Opinions on Interface Customization in World of Warcraft. *ACE*, pp. 198–213.
- Debeauvais, T., Nardi, B., Schiano, D., Ducheneaut, D., Yee, N., 2011. If you build it they might stay: retention mechanisms in World of Warcraft. In: *FDG '11 Proceedings of the 6th International Conference on Foundations of Digital Games*, pp. 180–187.
- DFC Intelligence, 2016. *Global forecast: PC game markets. Asia, North America, Europe 2016 July*.
- Dickey, M.D., 2011. World of Warcraft and the impact of game culture and play in an undergraduate game design course. *Comput. Educ.* 56, 200–209.
- Djaouti, D., Alvarez, J., Jessel, J.-P., Methel, G., 2008. Play, game, world: anatomy of a videogame. *Int. J. Intell. Games Simul.* 5 (1).
- Ducheneaut, N., et al., 2006. Building an MMO with mass appeal: a look at gameplay in world of warcraft. *Games Culture* 1 (4), 281–317.
- Ducheneaut, N., Wen, M.-H., Yee, N., Wadley, G., 2009. Body and mind: a study of avatar personalization in three virtual worlds. *CHI'2009*.
- Evans, G.L., 2013. A novice researcher's first walk through the maze of grounded theory: rationalization for classical grounded theory. In: *Grounded Theory Review*, 12. Sociology Press, pp. 37–55. June.
- Falstein, N., (2004) *The 400 project/* Retrieved March 1, 2014 from https://www.theinspiration.com/400_project.htm.
- Fullerton, T., 2008. *Game Design Workshop: A Playcentric Approach to Creating Innovative Games*. CRC Press, USA.
- Gonzales, S.J.L., et al., 2013. Playability: analyzing user experience in video games. *Behav. IT* 31 (10), 1033–1054.
- Gregor, S., 2006. The nature of theory in information systems. *MIS Quarterly* 30 (3), 611–642.
- Gregor, S., 2014. Theory - still king but needing a revolution!. *JIT* 29 (4), 337–340.
- Gross, S., et al., 2012. Studying social relations in MMOG play: an illustration of using ethnography to frame "Big Data". In: *17th ICCG*, pp. 167–174.
- Guiard, J.M., Malinverni, L., Parés Burgués, N., 2014. Narrative-based elicitation: orchestrating contributions from experts and children. *CHI Extended Abstr.* 1159–1164.
- Guizzardi, R.S.S., Franch, X., Guizzardi, G., 2012. Applying a foundational ontology to analyze means-end links in the i* framework. In: *RCIS*, pp. 1–11.
- Holopainen, J. *Foundation of gameplay*, blekinge Institute of technology Doctoral Dissertation Series No 2011:02, 2011.
- Kassab, M., 2009. *Non-functional Requirements – Modeling and Assessment*. VDM Publ.
- Kasurinen, et al., 2014. Is requirements engineering useless in game development? *REFSQ* 1–16.
- Keith, C., 2010. *Agile Game Development With Scrum*. Addison-Wesley.
- King, N., Horrocks, C., 2010. *Interviews in Qualitative Research*. Sage.
- Klevjer, R., 2012. Enter the avatar. The phenomenology of prosthetic telepresence in computer games. In: Fossheim, H., Mandt Larsen, T., Sageng, J.R. (Eds.), *The Philosophy of Computer Games*. Springer, London & New York, pp. 17–38.
- Kotonya, G., Sommerville, I., 1989. *Requirements Engineering: Processes and Techniques*. Wiley.
- Lauesen, S., 2002. *Software requirements: Styles and Techniques*. Addison-Wesley.
- LeBlanc, M., 2006. Tools for creating dramatic game dynamics. In: Salen, K., Zimmerman, E. (Eds.), *The Game Design Reader: Rules for Play Anthology*. MIT Press, pp. 438–459.

- Lopez, M.C., Fialho, F.A.P., Cunha, C.J.C.A., Viveiros, S.I., 2013. Business games for leadership development: a systematic review. *Simul. Gaming* 44 (4), 523–543.
- Mekler, E.D., Ayumi Bopp, J., Tuch, A.N., Opwis, K. A systematic review of quantitative studies on the enjoyment of digital entertainment games. *CHI 2014*: 927–936.
- Méndez Fernández, D., Penzenstadler, B., 2015. Artefact-based requirements engineering: the AMDiRE approach. *Requir. Eng.* 20 (4), 405–434.
- Mills, J., Bonner, A., Francis, K., 2006. The development of constructivist grounded theory. *Int. J. Qual. Methods* 5 (1), 25–36 Sage.
- Mueller, F., Gibbs, M.R., Vetere, F., Edge, D. Supporting the creative game design process with exertion cards. *CHI 2014*: 2211–2220.
- Murphy-Hill, E.R., Zimmermann, T., Nagappan, N., 2014. Cowboys, Ankle sprains, and Keepers of quality: How is Video Game Development Different from Software development?. *ICSE*, pp. 1–11.
- Nardi, B., 2010. *My Life as a Night Elf Priest*. Univ. of Michigan Press, Ann Arbor.
- Norris, N., 1997. Error, bias and validity in qualitative research. *Educ. Action Res.* 5 (1), 172–176.
- Nuseibeh, B., Easterbrook, S., 2000. Requirements engineering: a roadmap. In: *Proceedings of the Conference on the Future of Software Engineering*. ACM, pp. 35–46.
- Orita Almeida, M.S., Corrêa da Silva, F.S., 2013. A systematic review of game design methods and tools. In: *Entertainment Computing – ICEC 2013*, pp. 17–29. LNCS 8215.
- Östblad, P.A., Engström, H., Brusk, J., Backlund, P., 2014. Inclusive game design: audio interface in a graphical adventure game. In: *Proc. the 9th Audio Mostly (AM): A Conference on Interaction With Sound* Article No. 29.
- Paschali, M.E., Ampatzoglou, A., Chatzigeorgiou, A., Stamelos, I., 2014. Non-functional requirements that influence gaming experience: a survey on gamers satisfaction factors. *MindTrek* 208–215.
- Pasquale, L., et al., 2013. Requirements engineering meets physiotherapy: an experience with motion-based games. *REFSQ* 315–330.
- Pereira, L.L., Roque, L.A., 2013. Preliminary evaluation of a participation-centered gameplay experience design model. In: *SouthCHI*, pp. 332–348.
- Quax, P., Vanmontfort, W., Lamotte, W., Hautekeete, F., Vermeulen, P., Van Lier, K., 2014. A taxonomy A taxonomy for and analysis of the MMOG landscape. In: *Proc. 6th International Workshop on Massively Multiuser Virtual Environments*.
- Riekk, J., 2015. *Free-to-play Games: What are gamers paying for?* Bachelor's thesis. University of Oulu, Finland. <http://jultika.oulu.fi/files/nbnfioulu-201602271254.pdf>.
- Runeson, P., Höst, M., 2009. Guidelines for conducting and reporting case study research in software engineering. *ESE* 14 (2), 131–164.
- Scacchi, W., 2017. Practices and technologies in computer game software engineering. *IEEE Softw.* 34 (1), 110–116.
- Seddon, P., Scheepers, P., 2011. Towards the improved treatment of generalization of knowledge claims in IS research: drawing general conclusions from samples. *EJIS* 1–16.
- Sjøberg, D.I.K., Dybå, T., Jørgensen, M., 2007. The future of empirical methods in software engineering research. In: Briand, L.C., Wolf, A.L. (Eds.), *Int. Conf. on Software Engineering/Workshop on the Future of Software Engineering*. IEEE CS Press, pp. 358–378.
- Snyder, C., 2003. *Paper Prototyping: The Fast and Easy Way to Design and Refine User Interfaces*. Morgan Kaufmann.
- Sublette, V.A., Mullan, B., 2012. Consequences of play: a systematic review of the effects of online gaming. *Int. J. Mental Health* 10, 3–23.
- Sullivan, A., Smith, G., 2011. Lessons from teaching game design. In: *Proc. of the 6th Int. Conference on Foundations of Digital Games (FDG '11)*. ACM, pp. 307–309.
- Teruel, M.Á., 2017. *Improving Collaborative Post-WIMP Systems through Requirements Specification* PhD Thesis. University of Castilla-La Mancha, Albacete, Spain. <http://hdl.handle.net/10578/12482>.
- Teruel, M.A., Navarro, E., González, P., López-Jaquero, V., Simarro, F.M., 2016. Applying thematic analysis to define an awareness interpretation for collaborative computer games. *Inf. Softw. Technol.* 74, 17–44.
- Valente, L., Feijó, B., Sampaio do Prado Leite, J.C., 2017. Mapping quality requirements for pervasive mobile games. *Requir. Eng.* 22 (1), 137–165.
- Verganti, V., Pisano, G., 2008. Which kind of collaboration is right for you? *Harvard Bus. Rev.* 86 (12) Dec.
- Warmelink, H., Siitonen, M., 2011. Player Communities in Multiplayer Online Games: A Systematic Review. *DiGRA*, pp. 1–21.
- Williams, J.P., Kirschner, D., Suhaimi-Broder, Z., 2014. Structural roles in massively multiplayer online games: a case study of guild and raid leaders in world of warcraft. *Symbolic Interact. New Media* 43, 121–142.
- Yin, R.K., 2008. *Case Study Research*. Sage.
- Zagal, P.J., Bruckman, A., 2008. The game ontology project: supporting learning while contributing authentically to game studies. *ICLS* (2) 499–506.
- Zhu, M., Zhao, F., Fang, X., Moser, C., 2017. Developing playability heuristics based on nouns and adjectives from online game reviews. *Int. J. Hum.-Comput. Interact.* 33 (3), 241–253 2017.

Maya Daneva, PhD, is Senior Member of Scientific Staff in the Services, Cybersecurity and Safety group at the University of Twente, the Netherlands. Her key research interests are requirements engineering for large systems, players-game interaction, game analytics, crowdsourcing-based requirements engineering, requirements-based project estimation, and empirical research methods. Maya has a strong international exposure having spent two years of her career in Germany at the University of Saarbruecken and in the IDS Scheer, and 9 years as a business process analyst for SAP Enterprise Recourse Planning projects at TELUS Corporation, Canada's second largest telecommunication company. At Twente, she is a leading researcher in seven industry-university research projects. Maya serves as the University of Twente's representative to ISERN, the International Empirical Software Engineering Research Network. She has been a Member of the Editorial Board of the Information and Software Technology Journal and a program chair of the 2016 International Conference of Requirements Engineering – Foundation for Software Quality (REFSQ).